

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

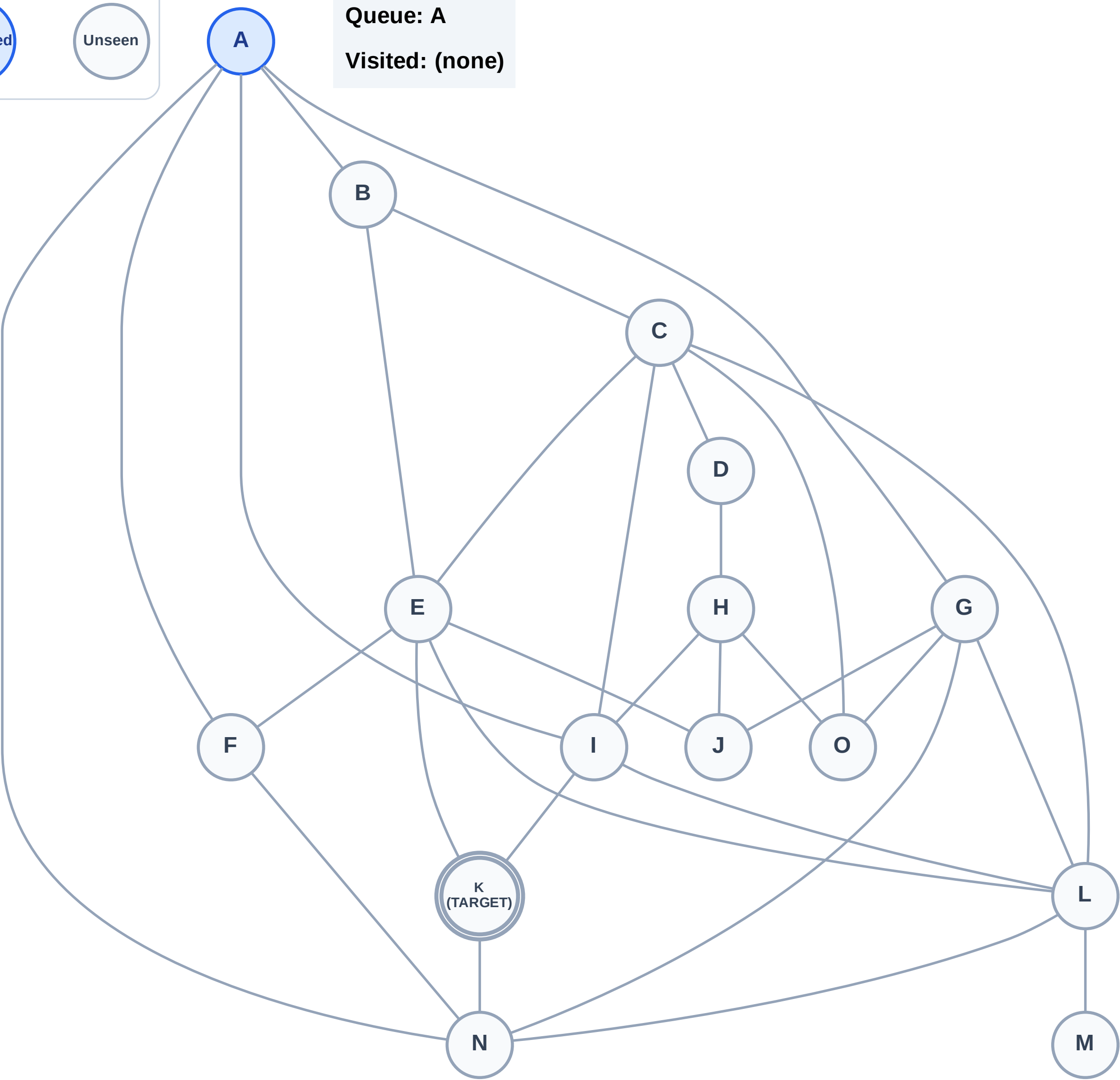
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

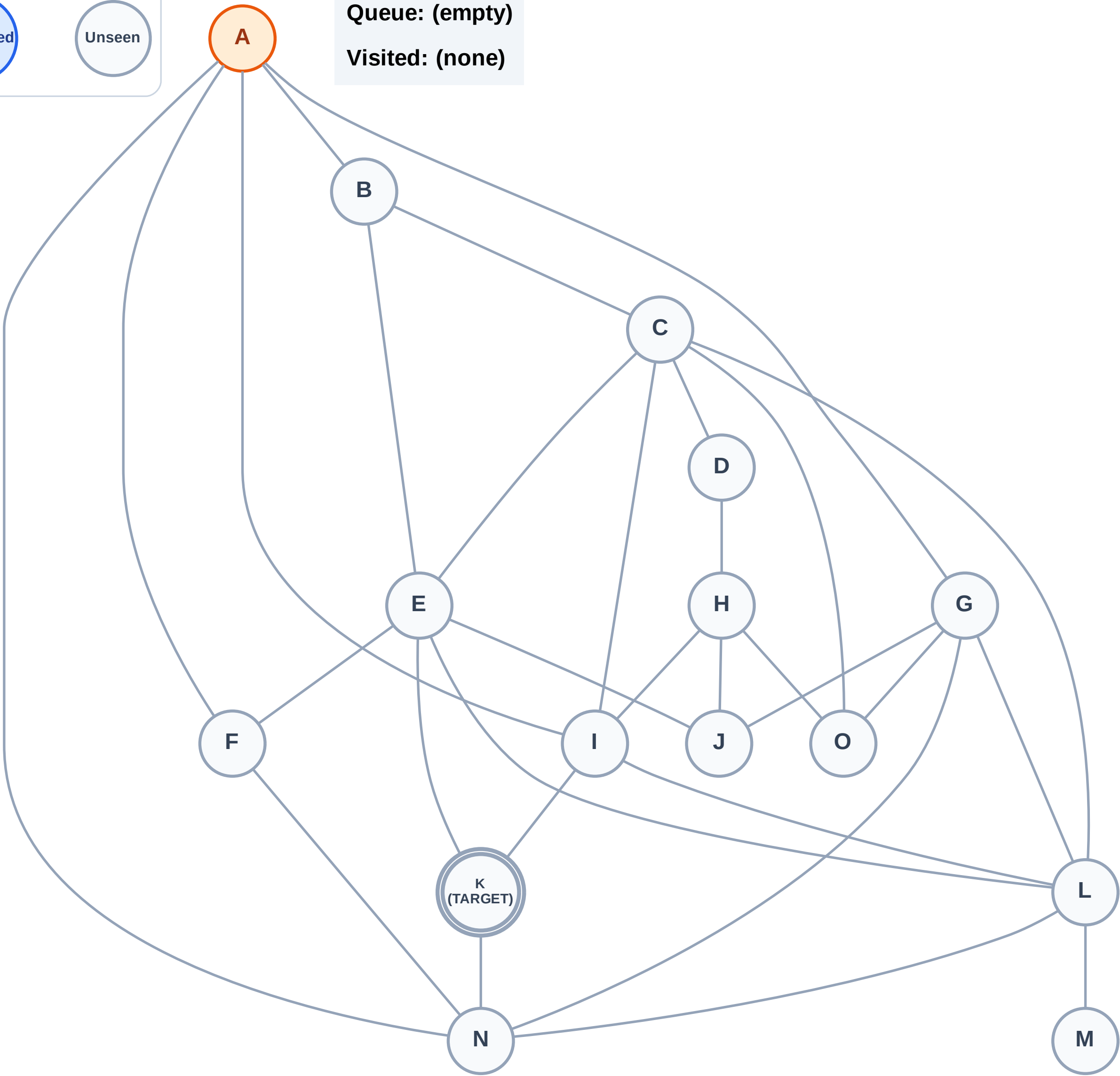
Complete

Processing

Discovered

Unseen

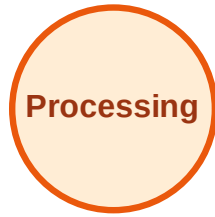
Queue: (empty)
Visited: (none)



BFS Traversal

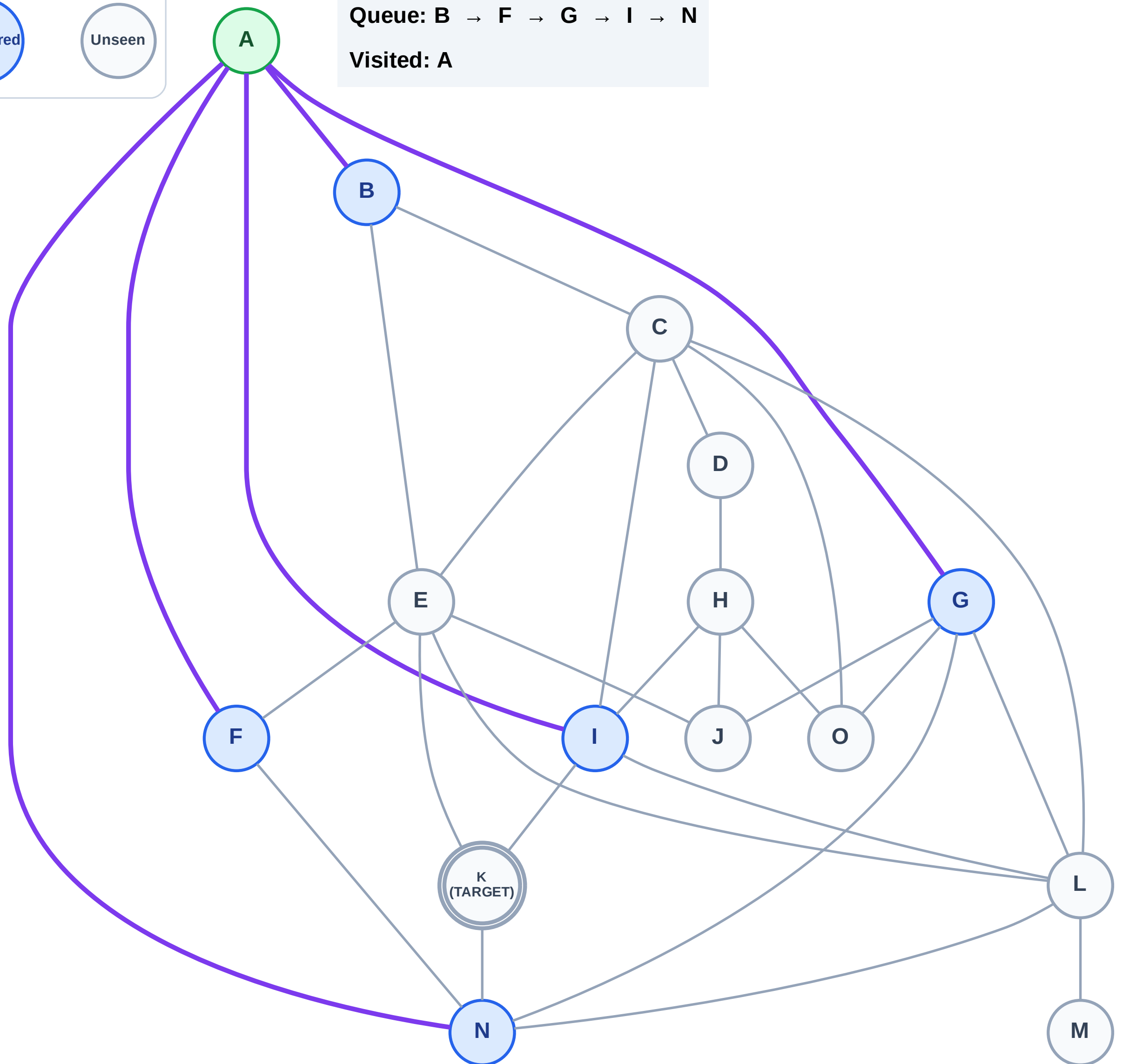
Step 3: Discover B, F, G, I, N from A

Legend



Queue: B → F → G → I → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

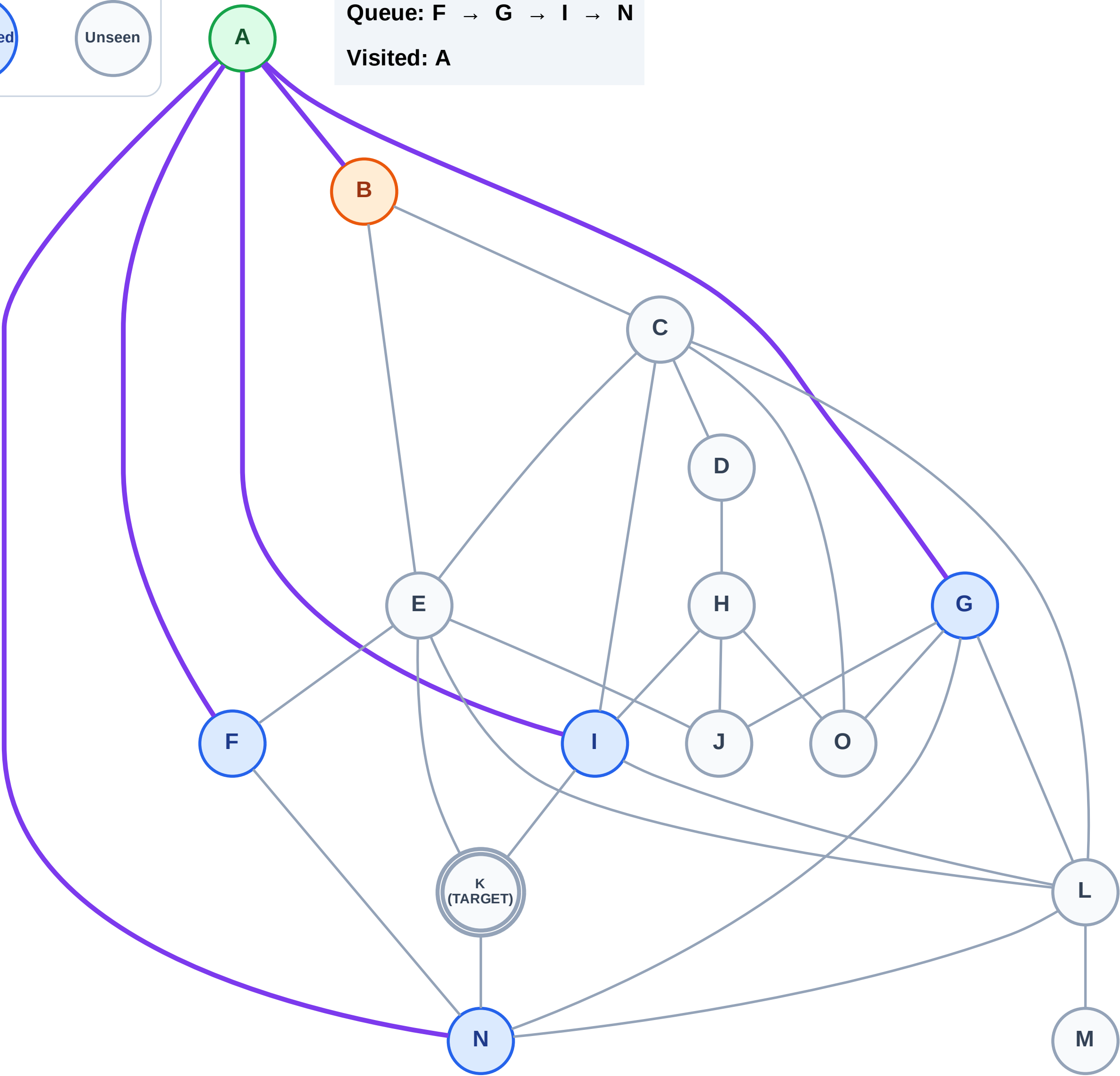
Processing

Discovered

Unseen

Queue: F → G → I → N

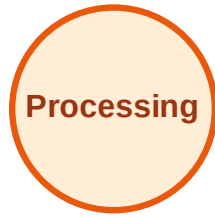
Visited: A



BFS Traversal

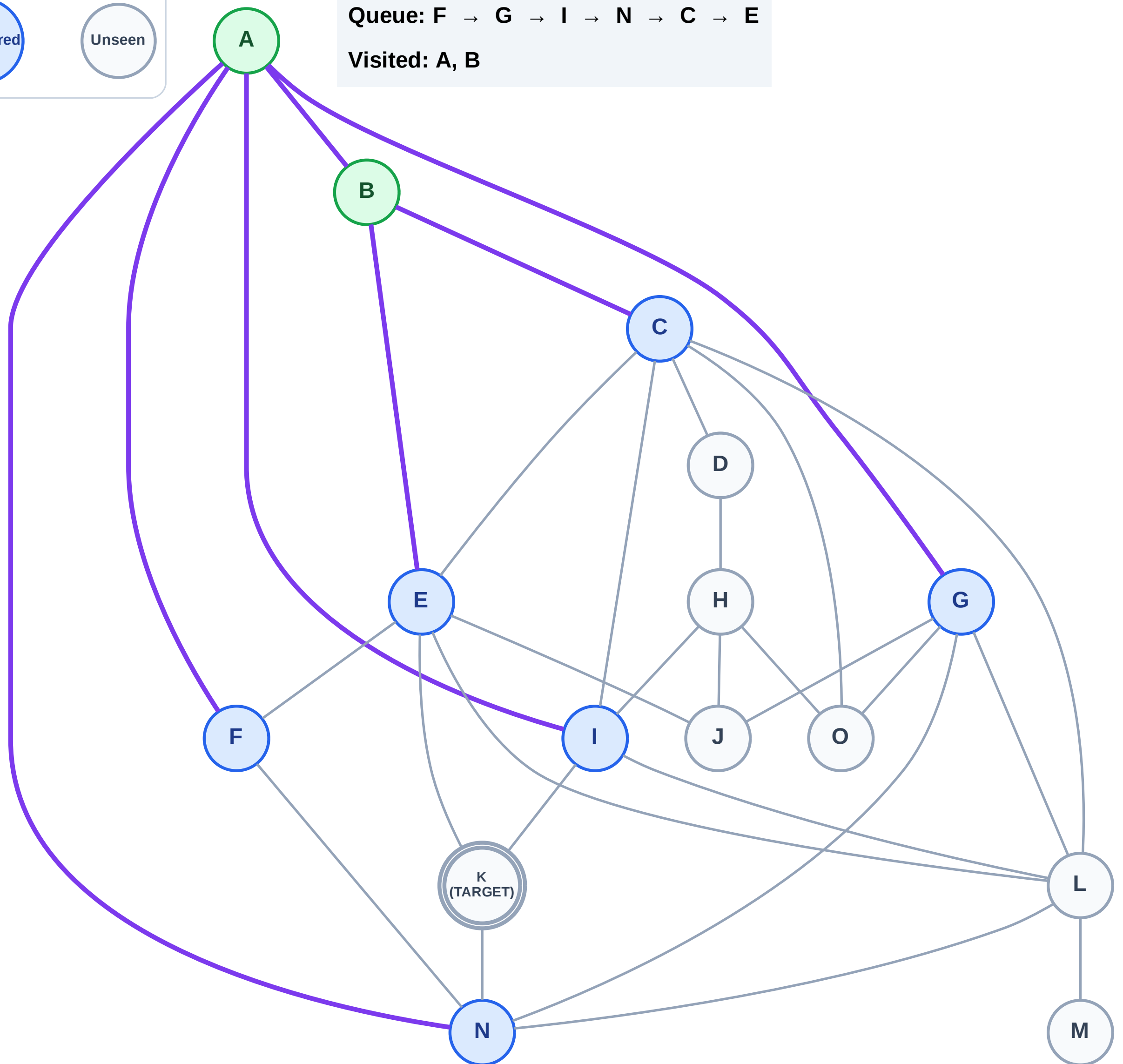
Step 5: Discover C, E from B

Legend



Queue: F → G → I → N → C → E

Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

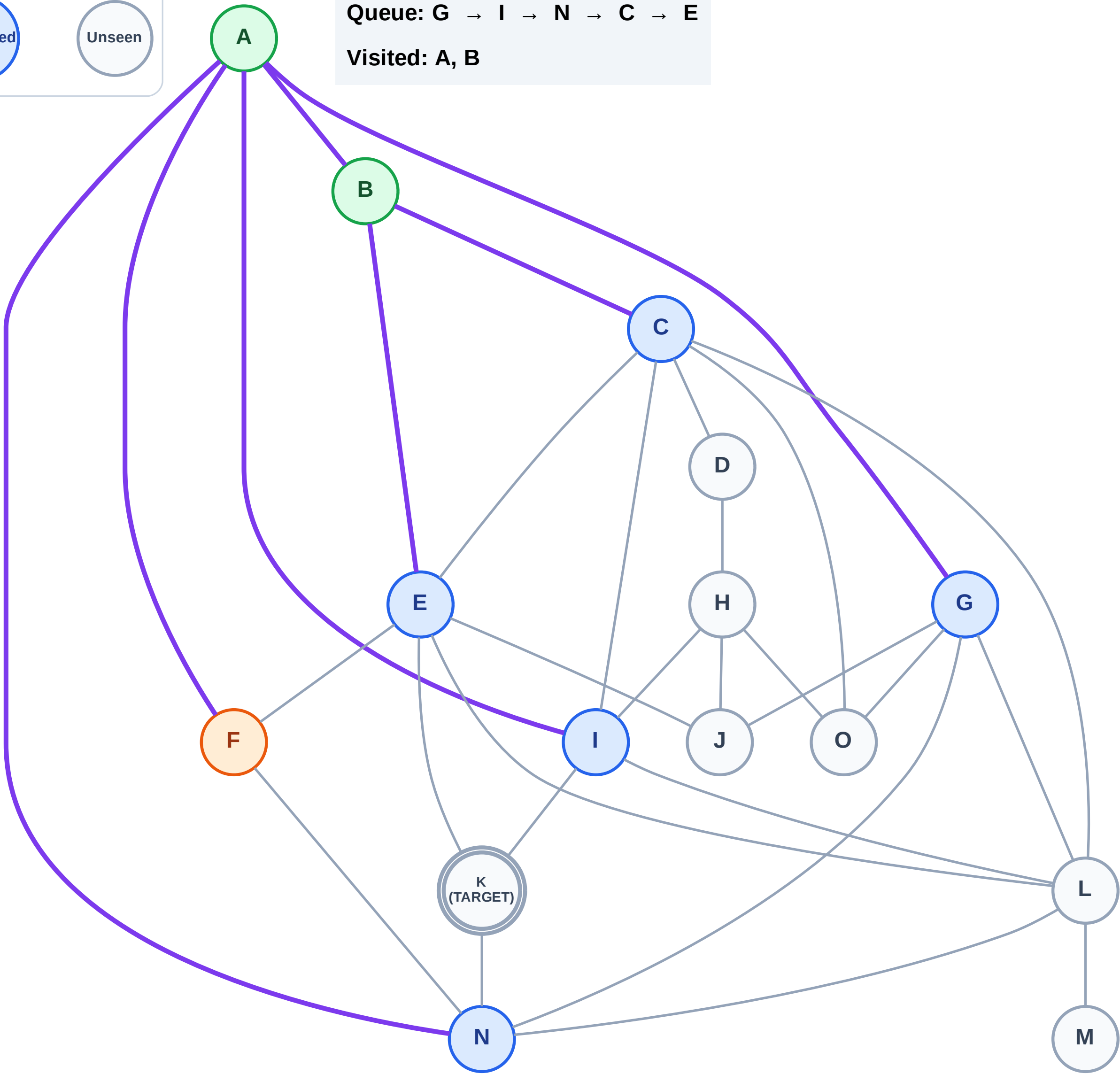
Processing

Discovered

Unseen

Queue: G → I → N → C → E

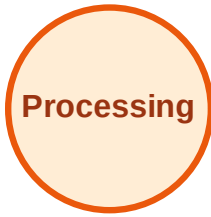
Visited: A, B



BFS Traversal

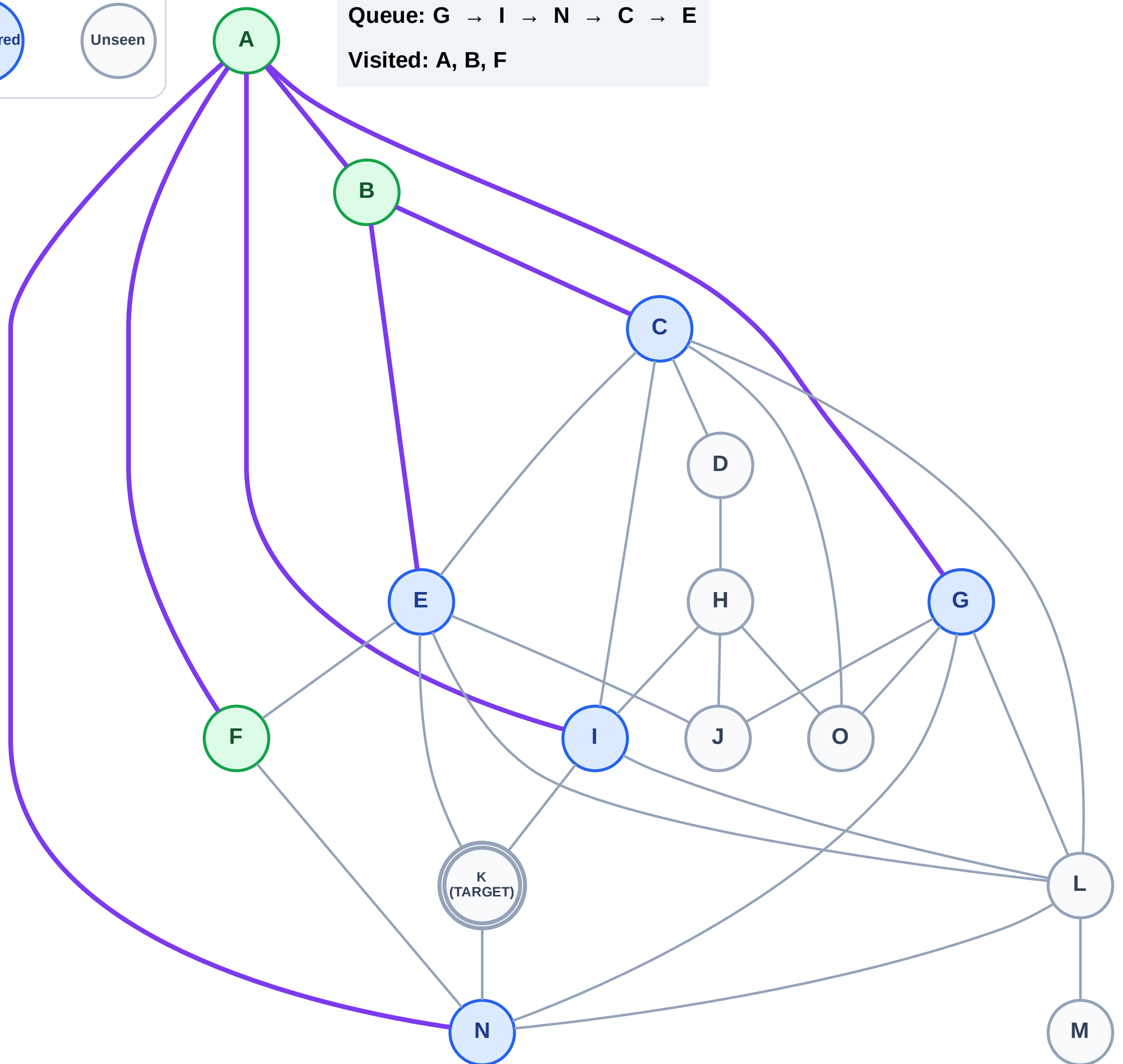
Step 7: No new neighbors from F

Legend



Queue: G → I → N → C → E

Visited: A, B, F



BFS Traversal

Step 8: Process node G

Legend

Complete

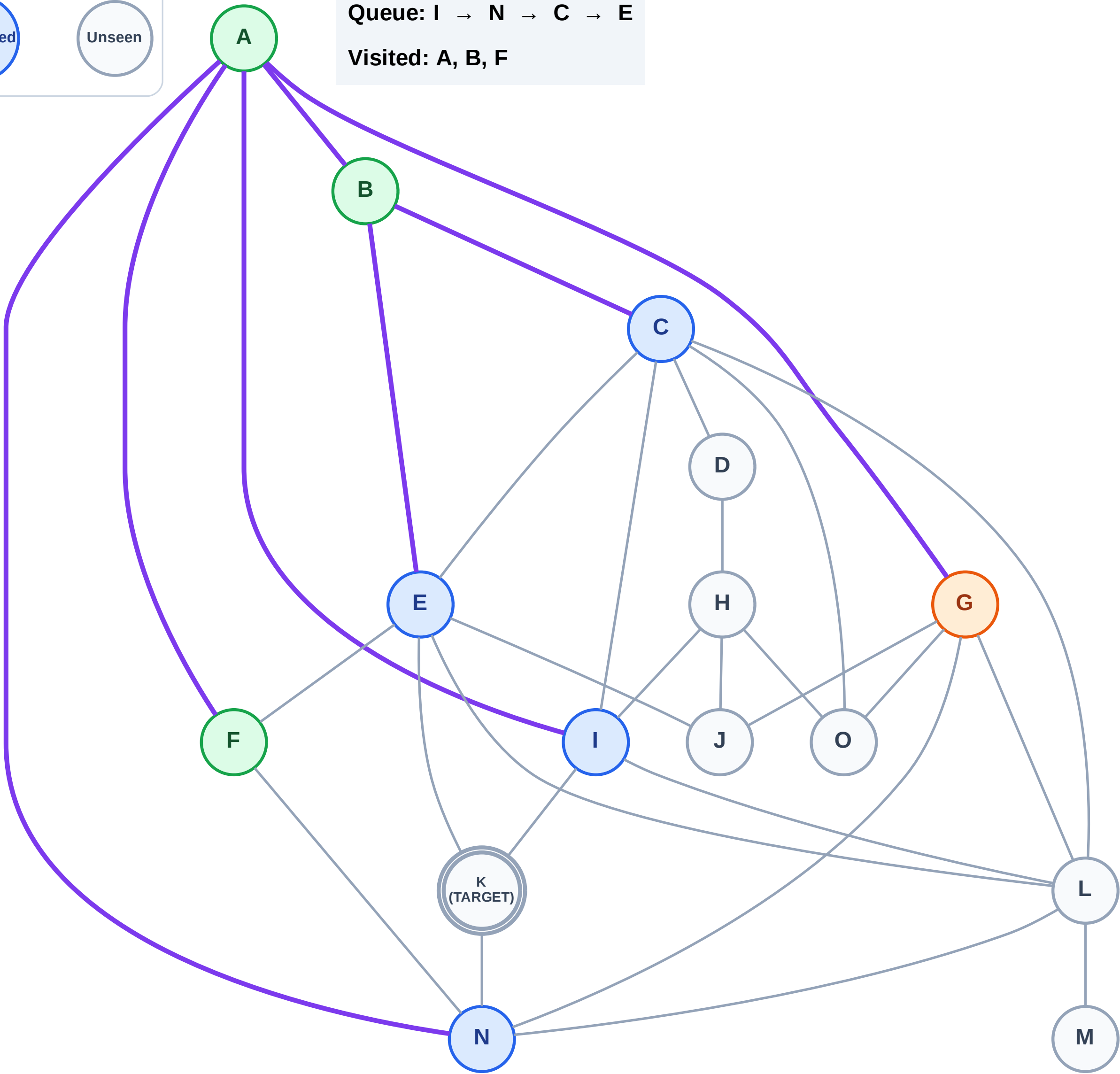
Processing

Discovered

Unseen

Queue: I → N → C → E

Visited: A, B, F



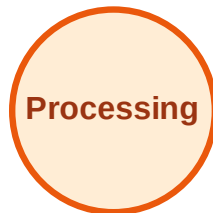
BFS Traversal

Step 9: Discover J, L, O from G

Legend



Complete



Processing



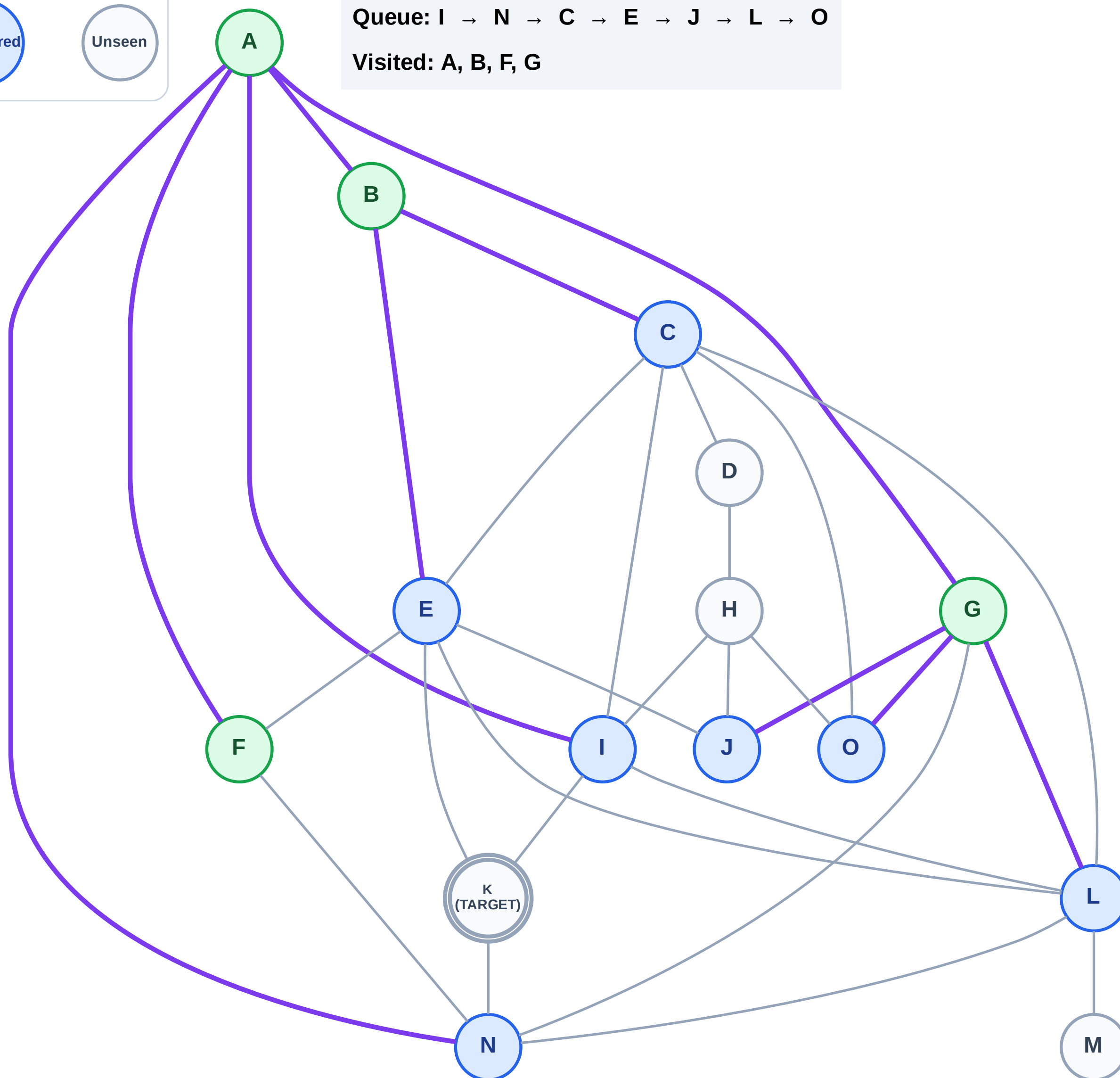
Discovered



Unseen

Queue: I → N → C → E → J → L → O

Visited: A, B, F, G



BFS Traversal

Step 10: Process node I

Legend

Complete

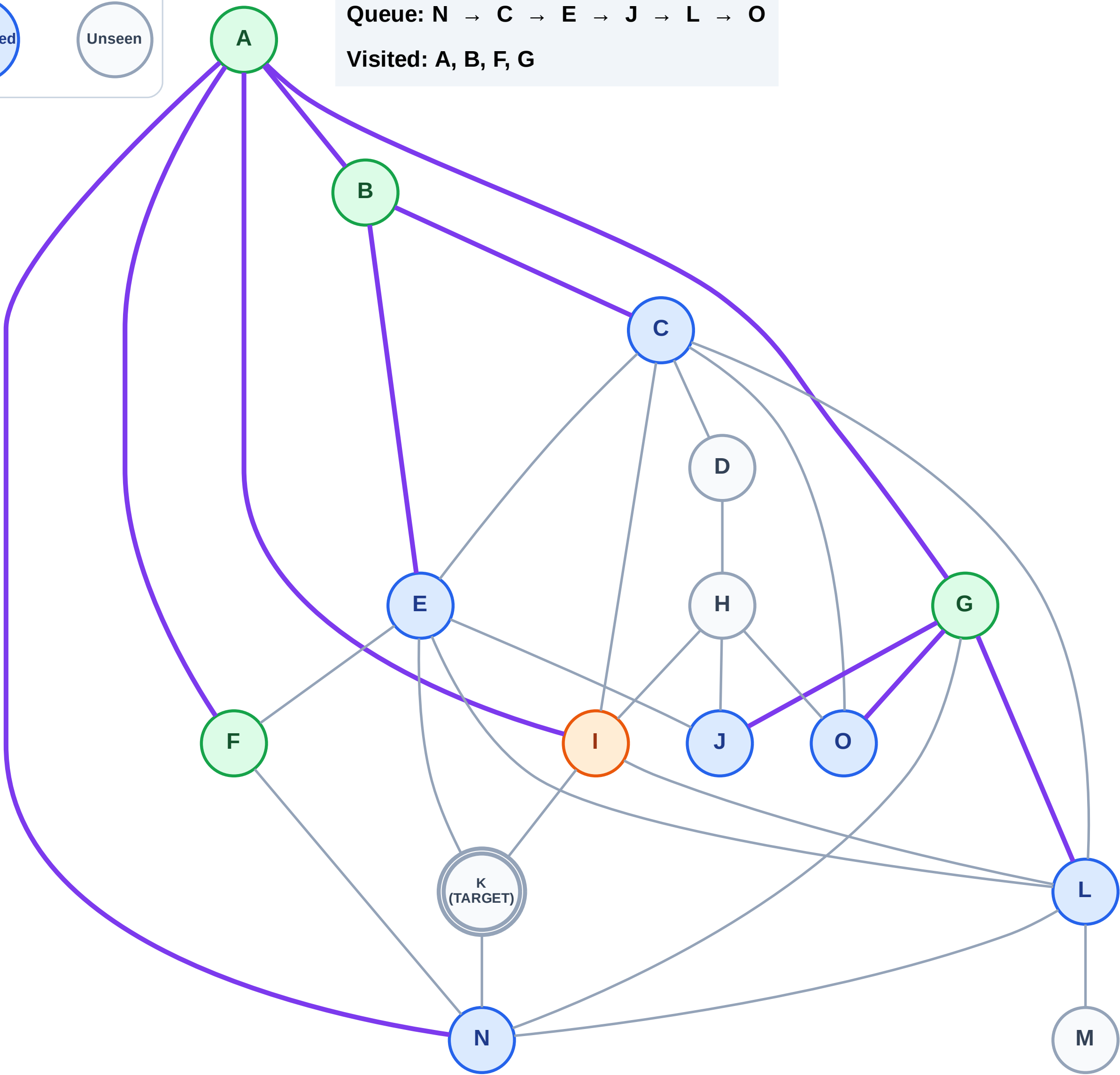
Processing

Discovered

Unseen

Queue: N → C → E → J → L → O

Visited: A, B, F, G



BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → I → K

Legend

Complete

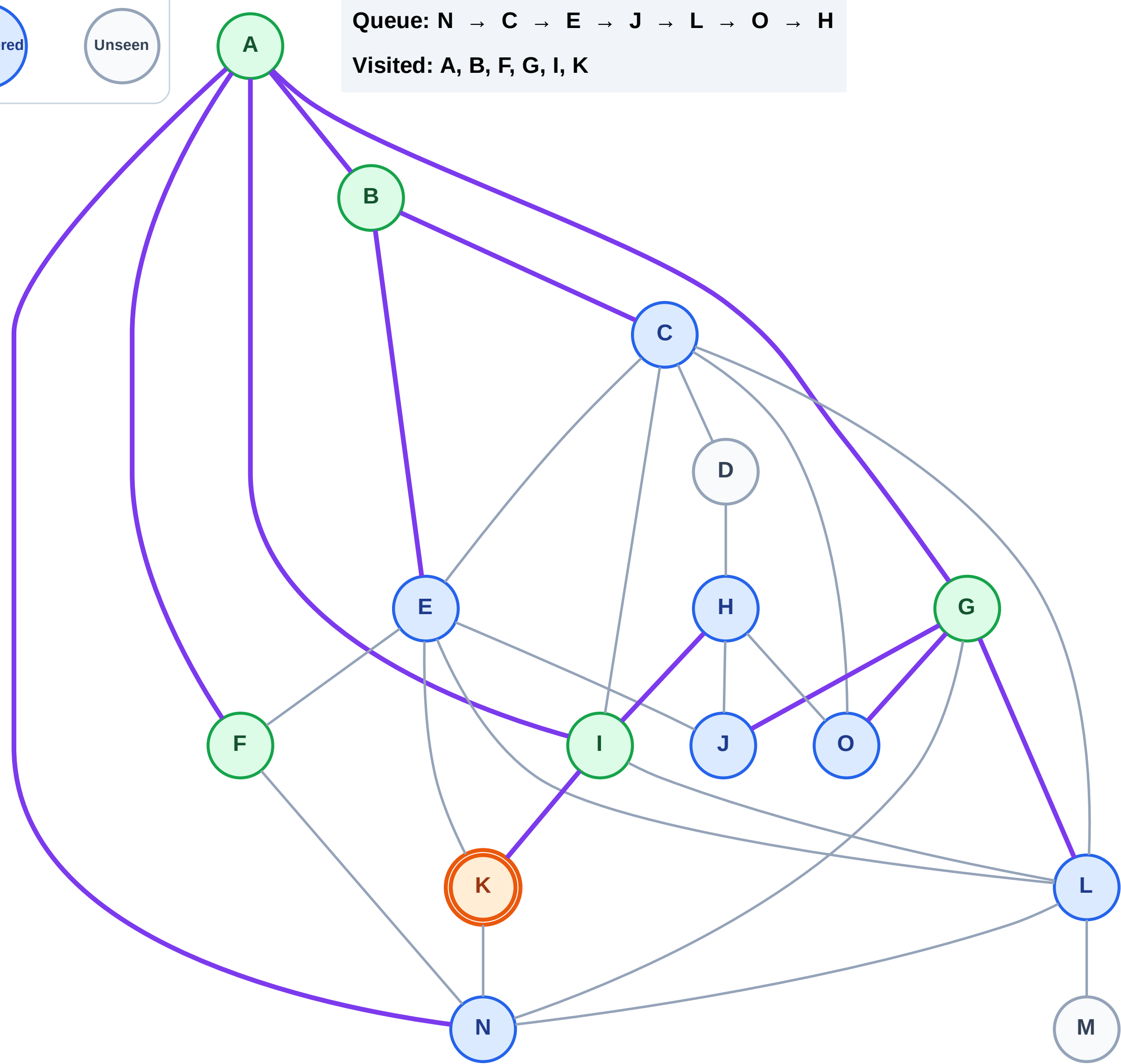
Processing

Discovered

Unseen

Queue: N → C → E → J → L → O → H

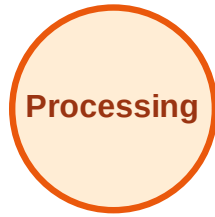
Visited: A, B, F, G, I, K



DFS Traversal

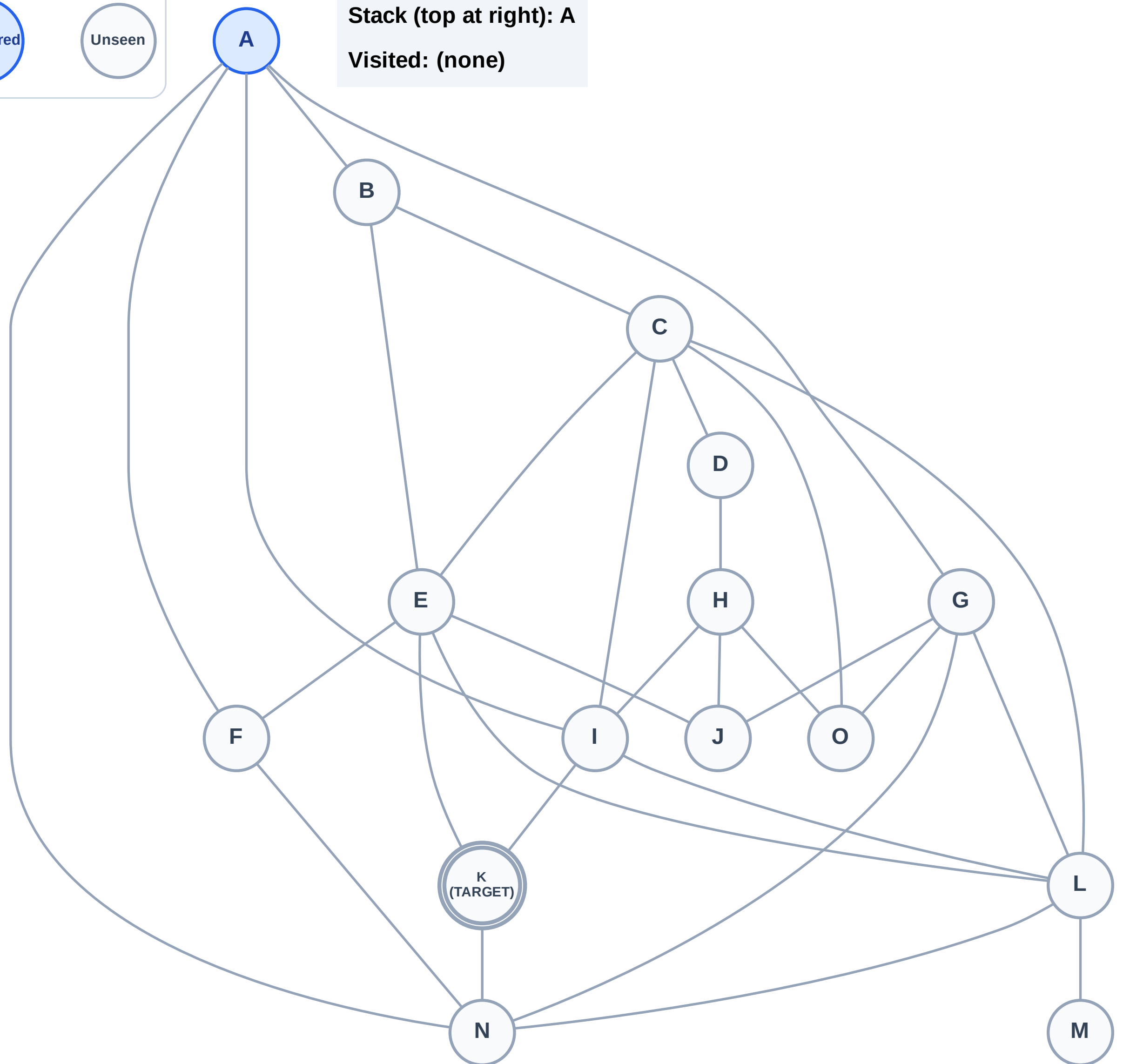
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

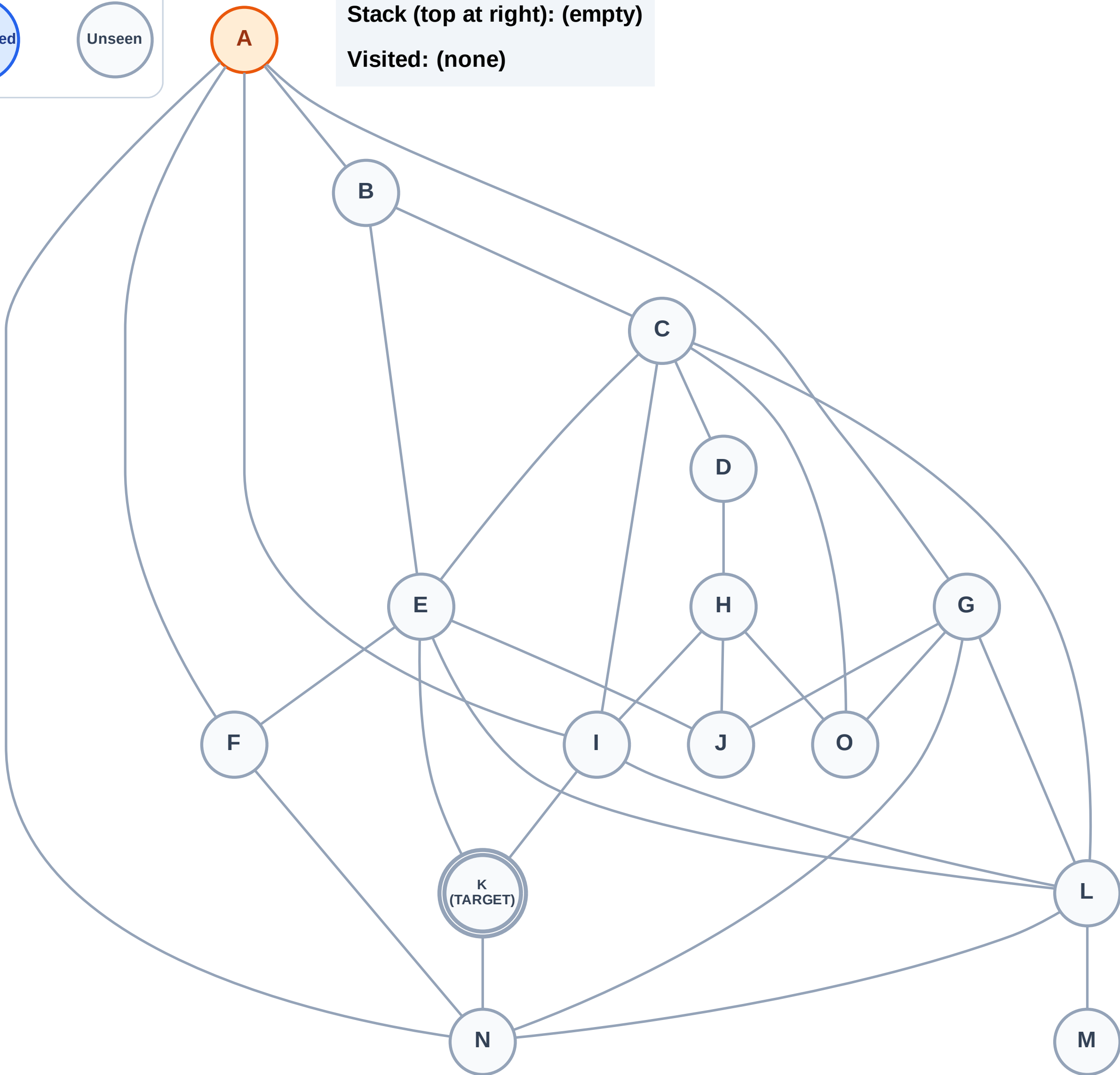
Complete

Processing

Discovered

Unseen

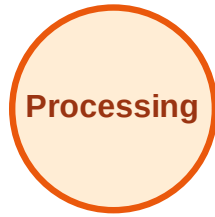
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

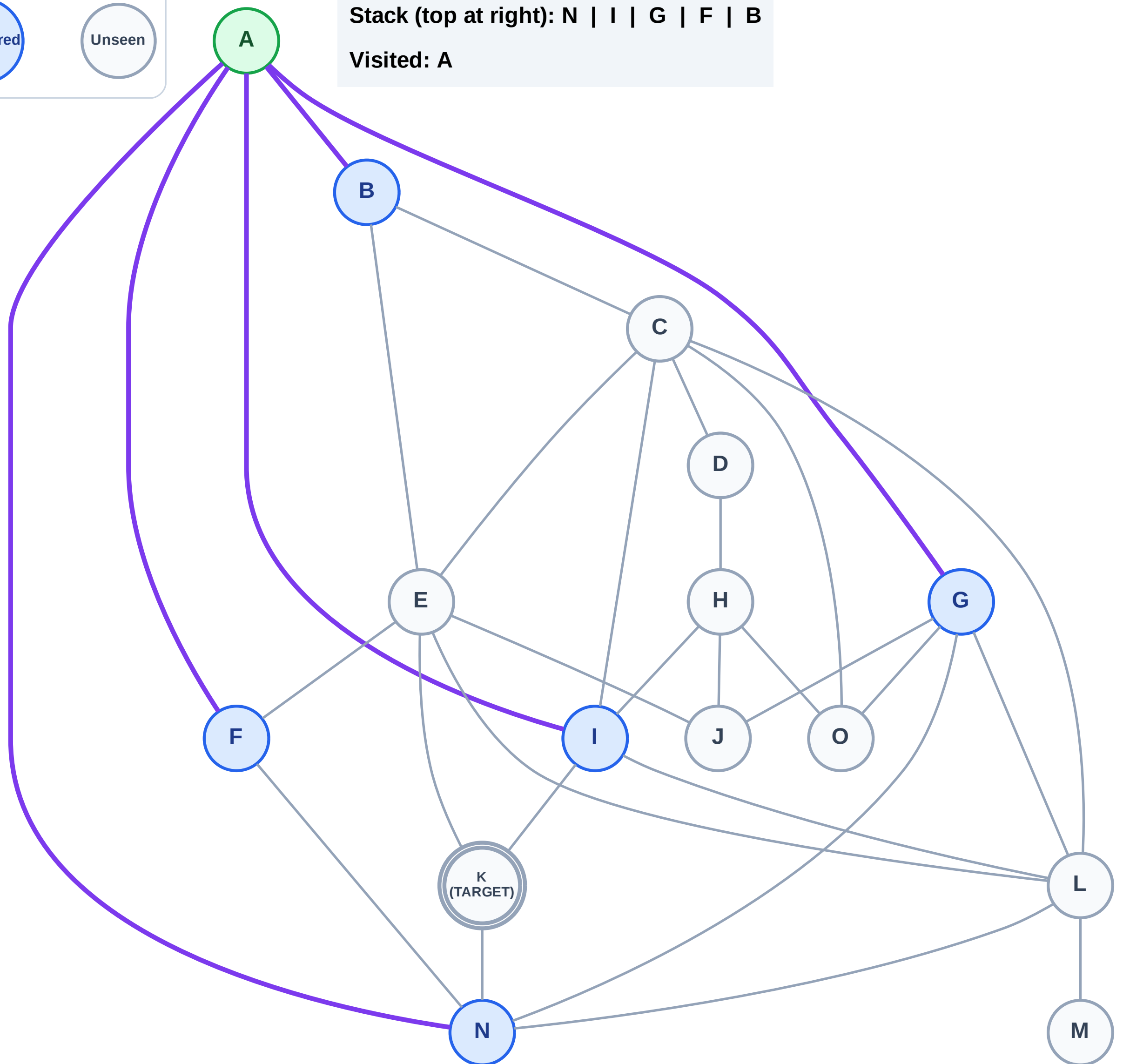
Step 3: Discover N, I, G, F, B from A

Legend



Stack (top at right): N | I | G | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

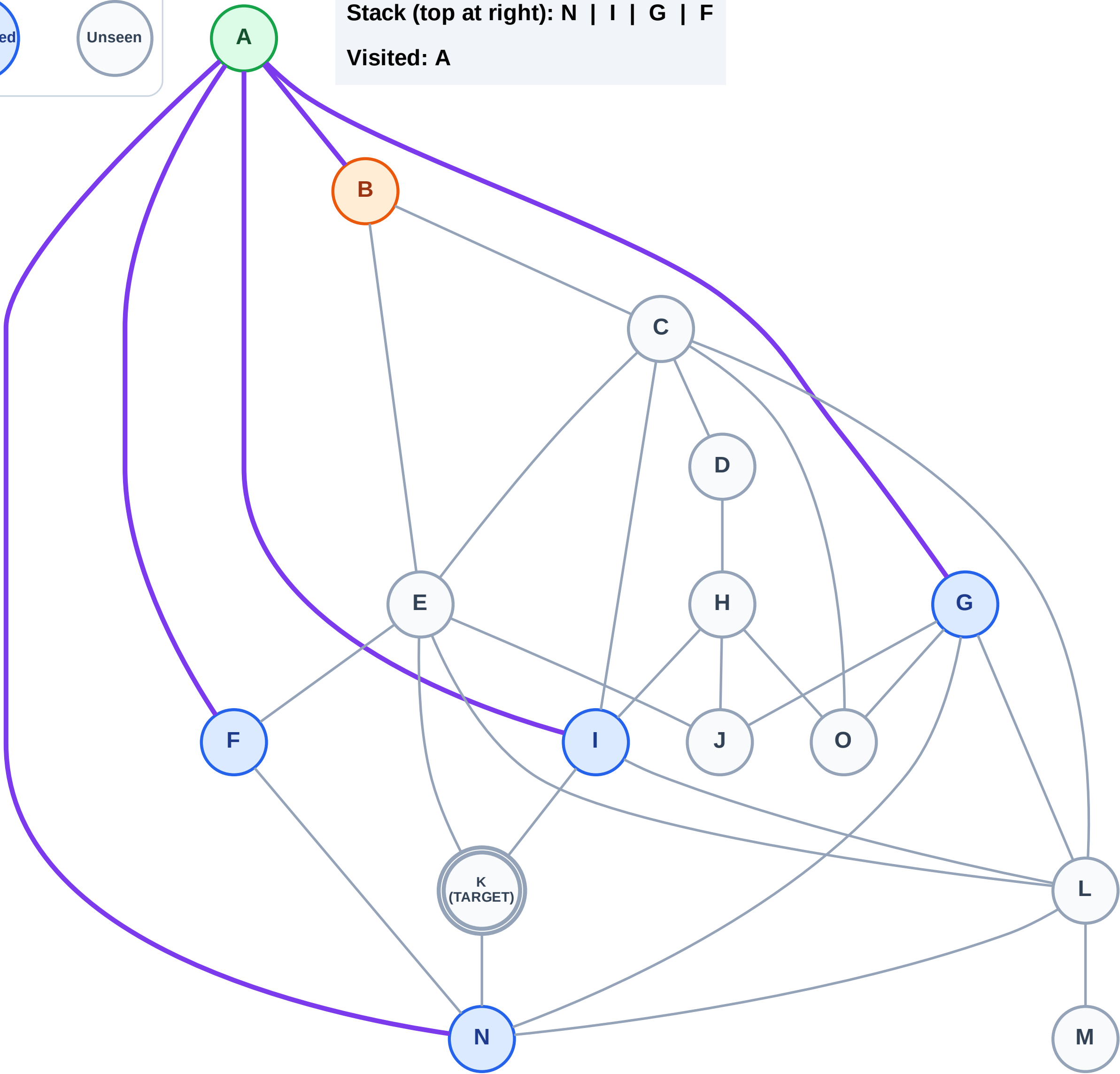
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F

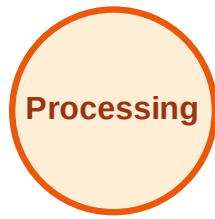
Visited: A



DFS Traversal

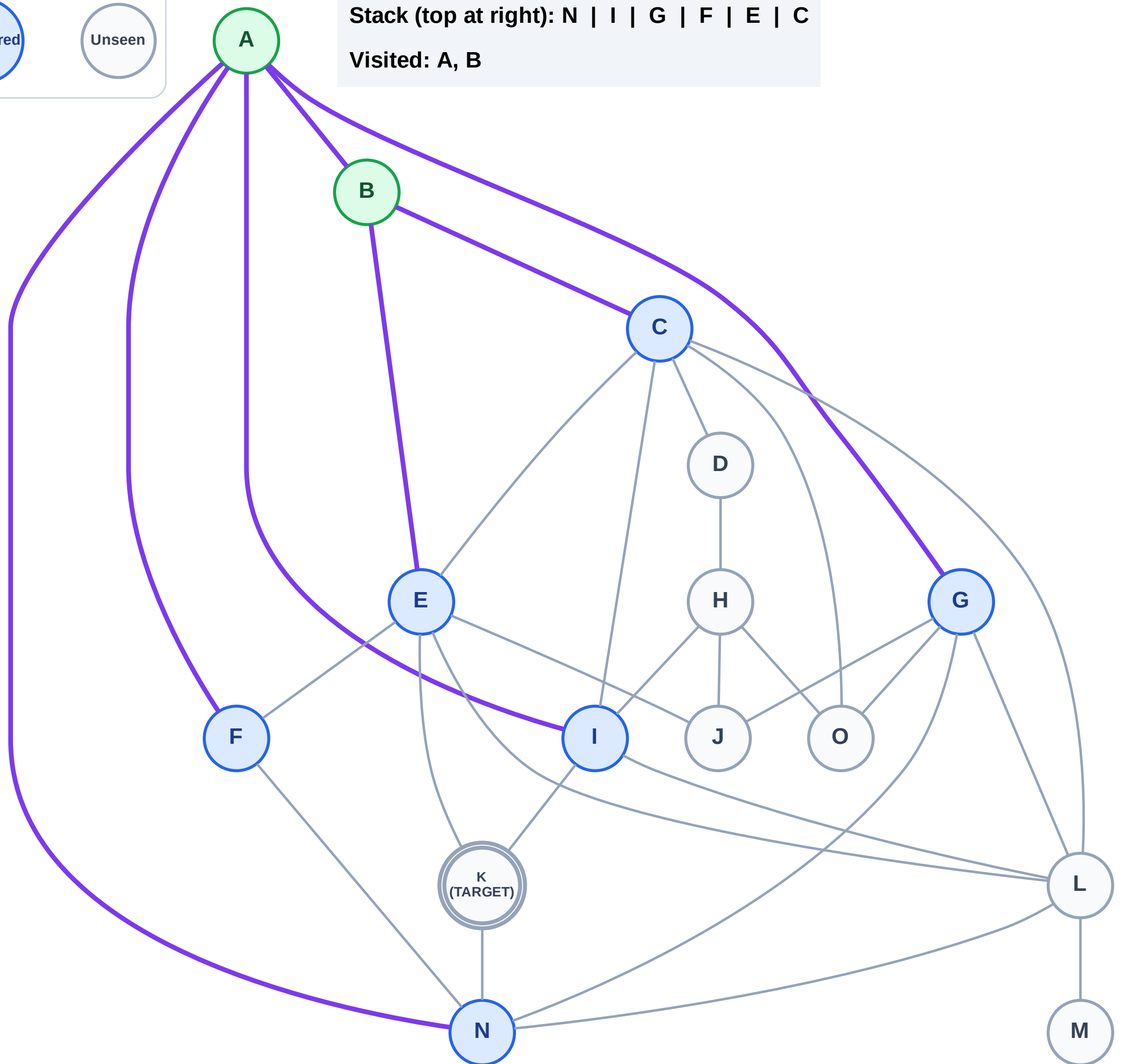
Step 5: Discover E, C from B

Legend



Stack (top at right): N | I | G | F | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

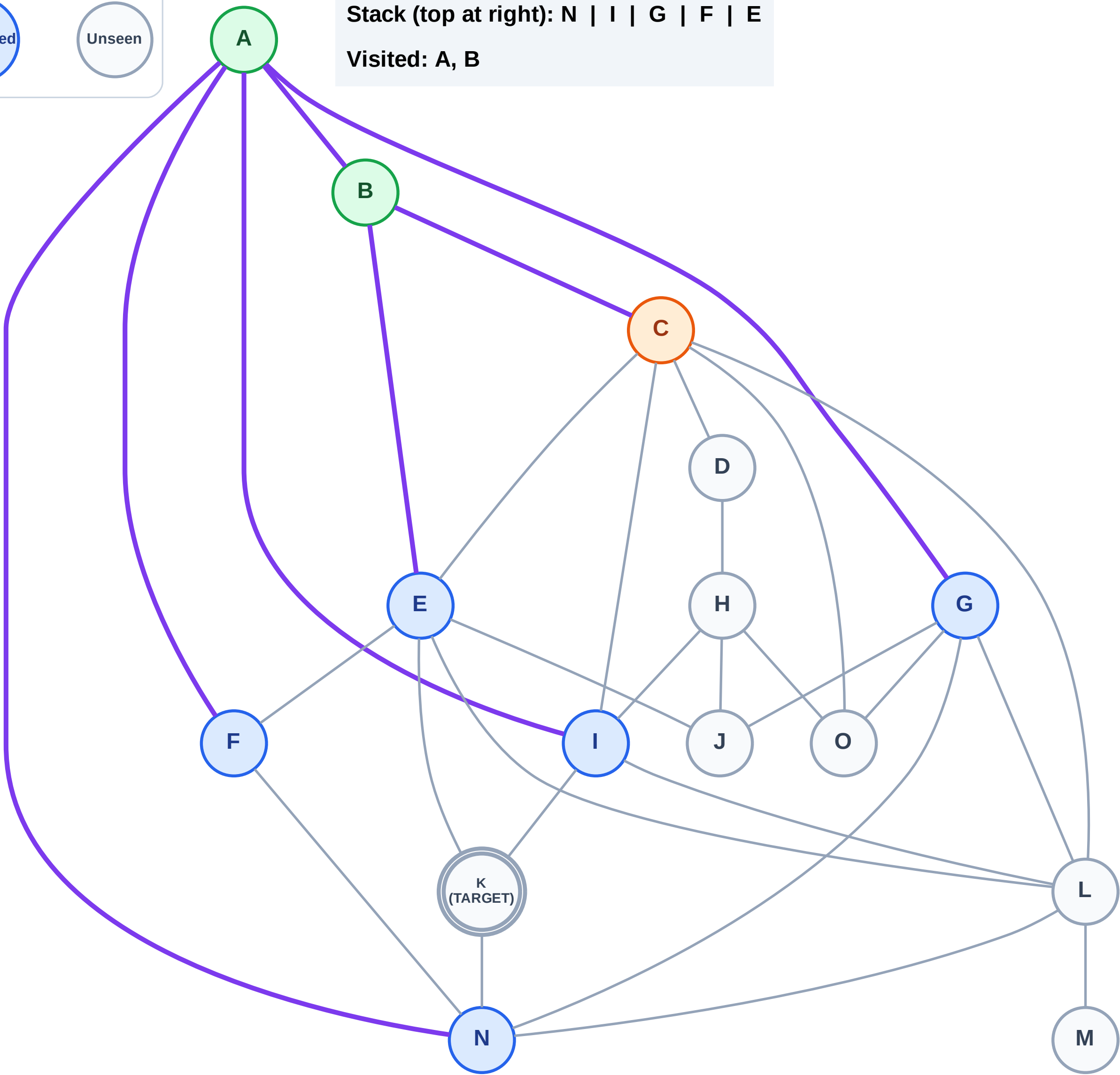
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F | E

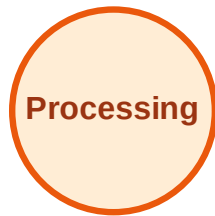
Visited: A, B



DFS Traversal

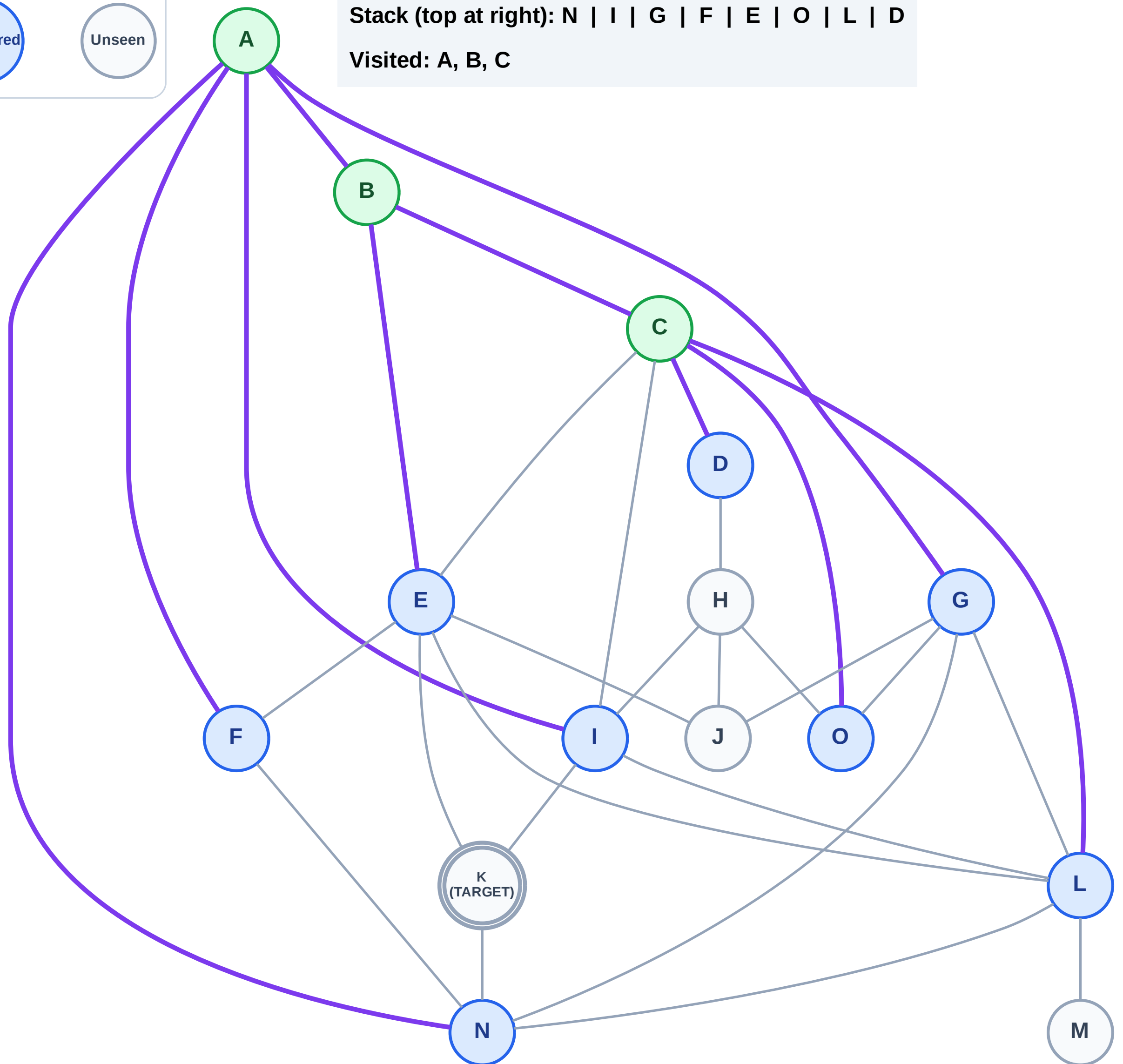
Step 7: Discover O, L, D from C

Legend



Stack (top at right): N | I | G | F | E | O | L | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

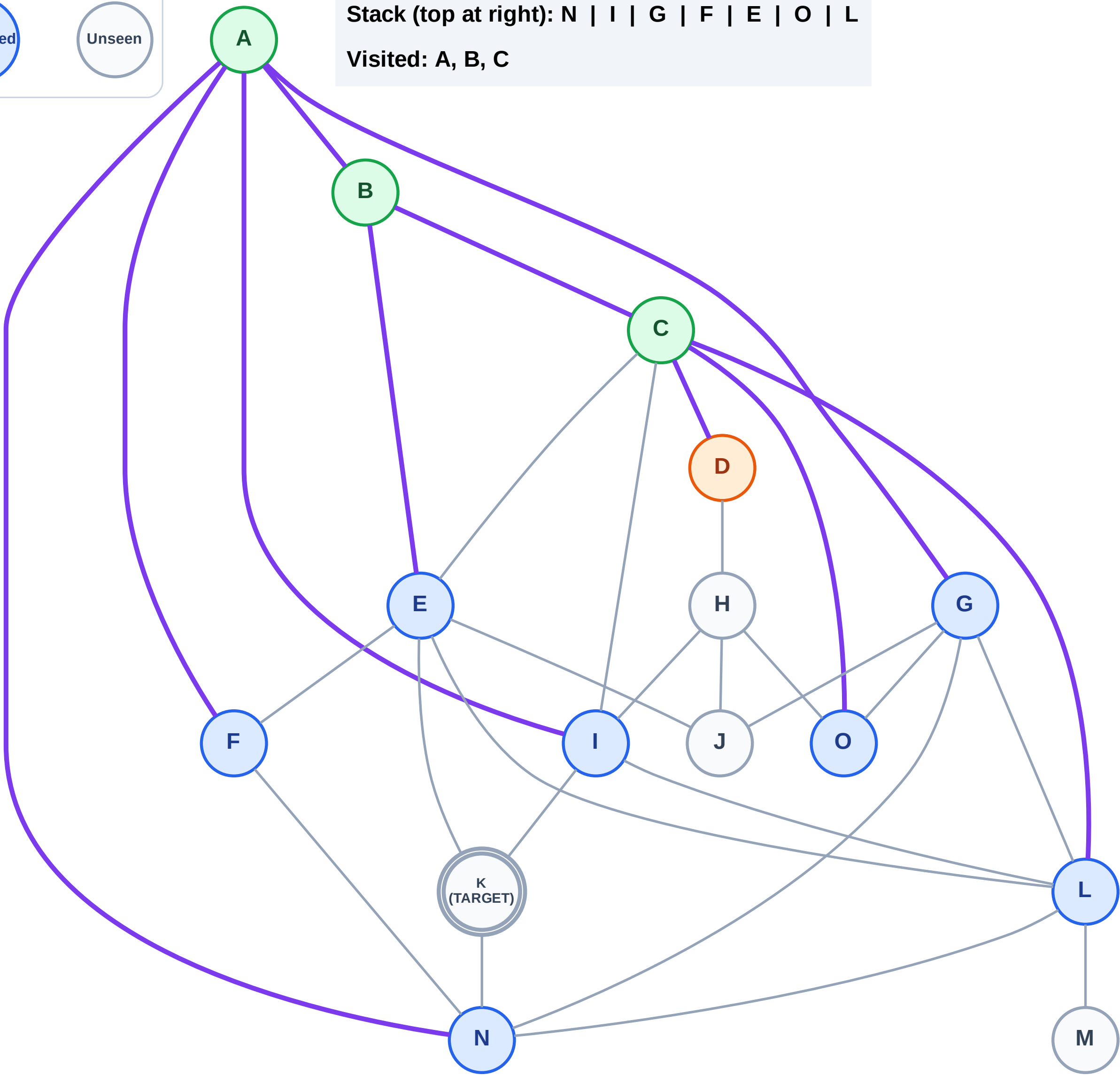
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F | E | O | L

Visited: A, B, C



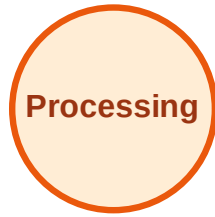
DFS Traversal

Step 9: Discover H from D

Legend



Complete



Processing



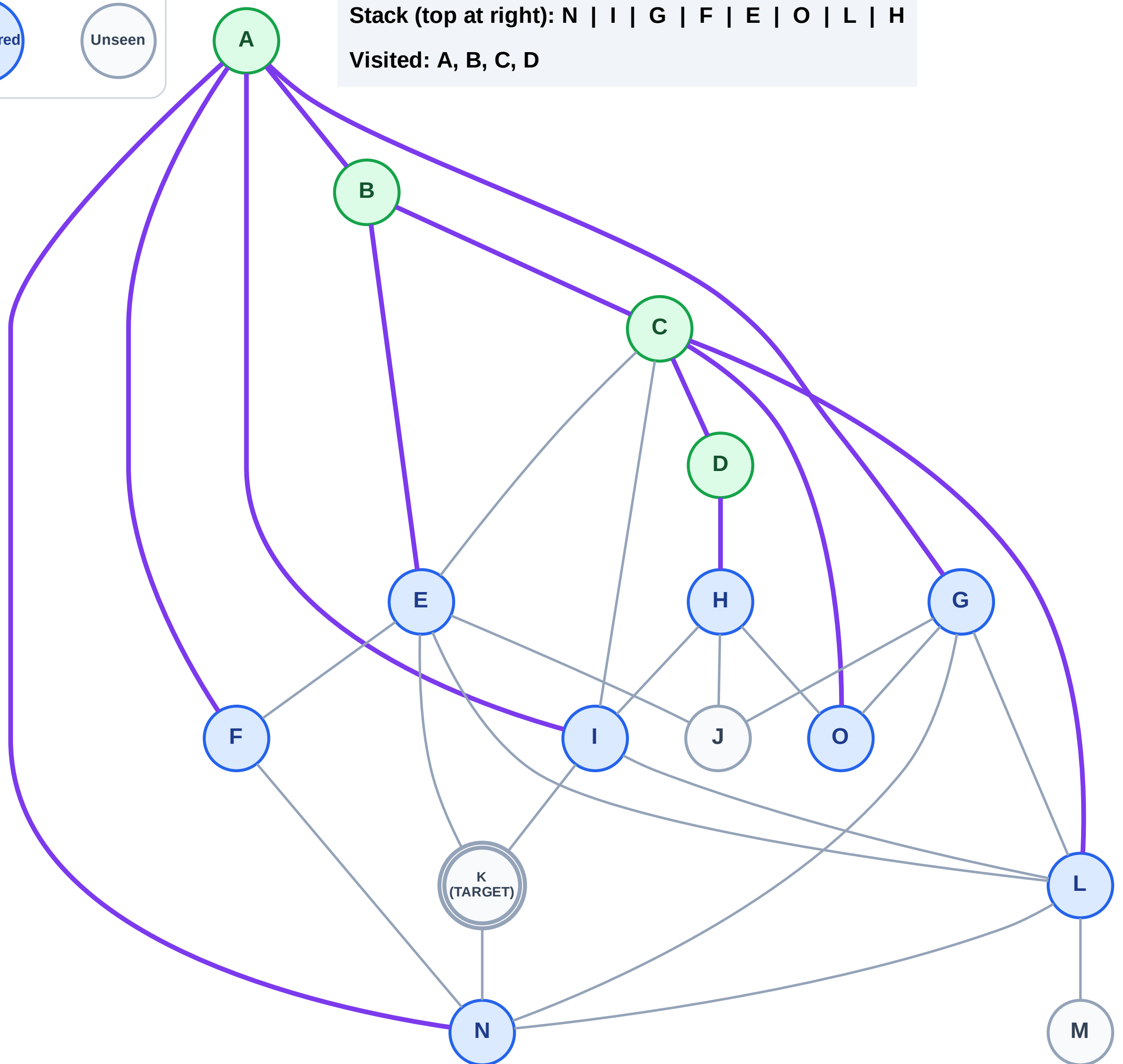
Discovered



Unseen

Stack (top at right): N | I | G | F | E | O | L | H

Visited: A, B, C, D



DFS Traversal

Step 10: Process node H

Legend

Complete

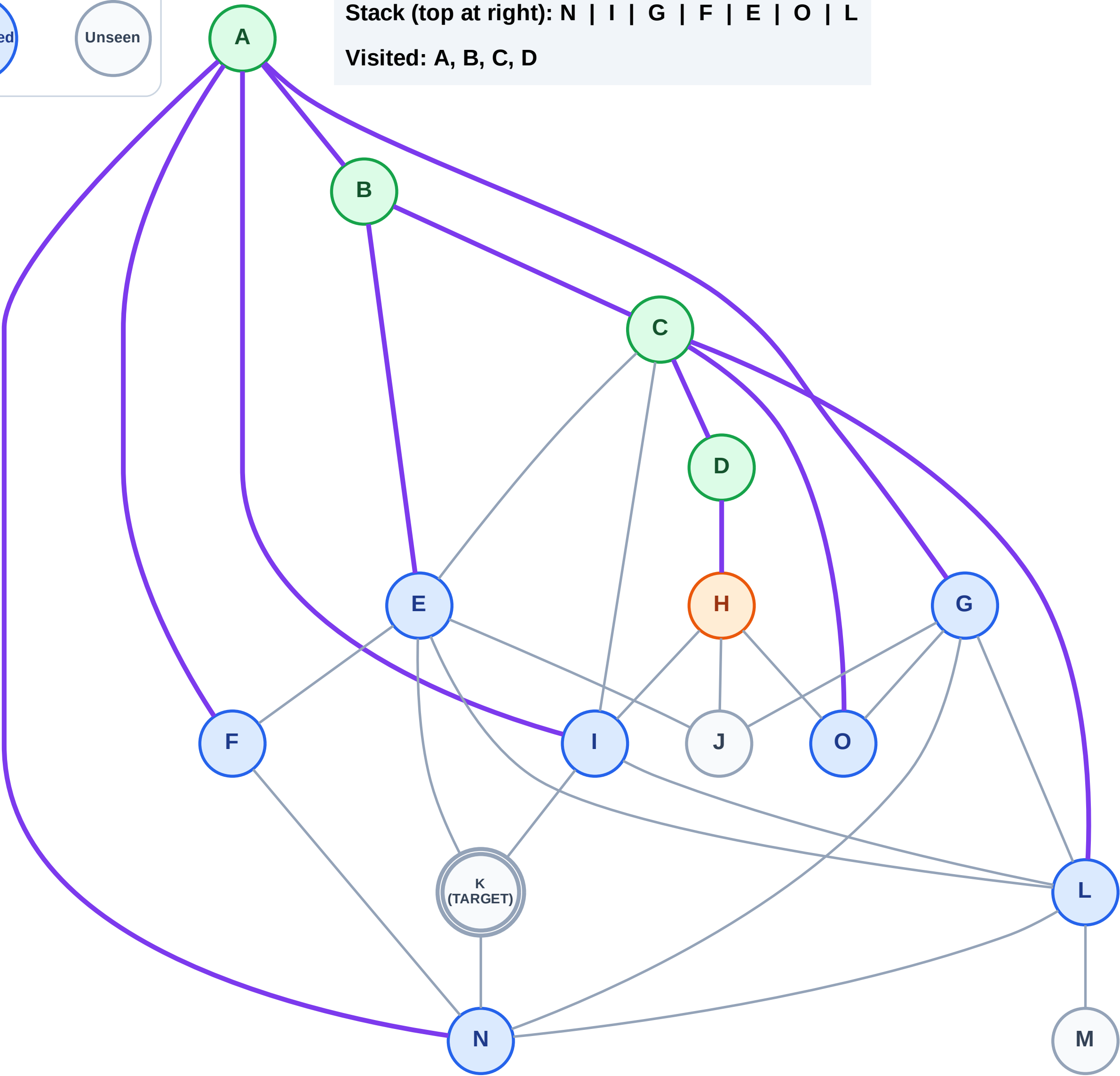
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F | E | O | L

Visited: A, B, C, D



DFS Traversal

Step 11: Discover J from H

Legend

Complete

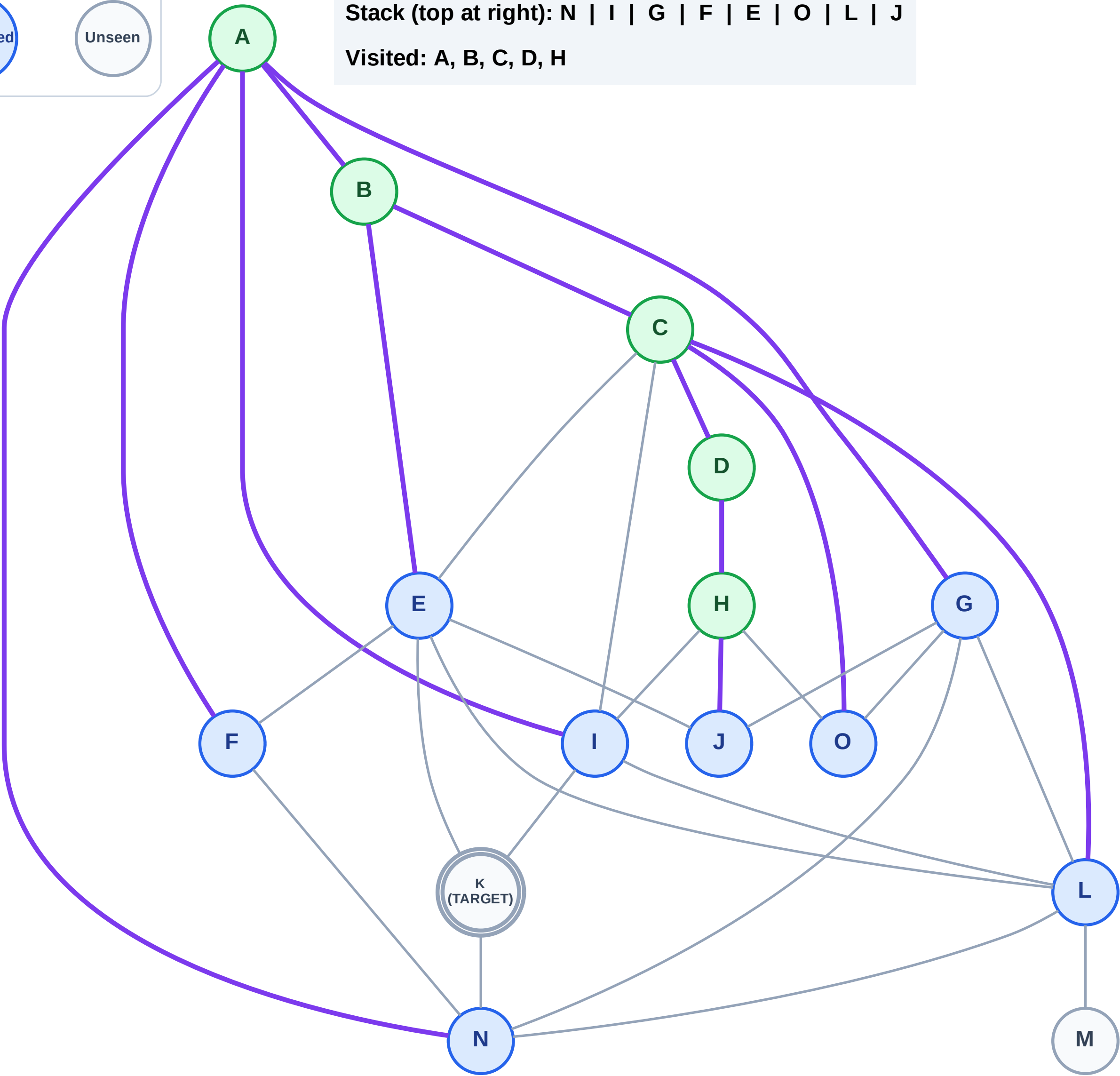
Processing

Discovered

Unseen

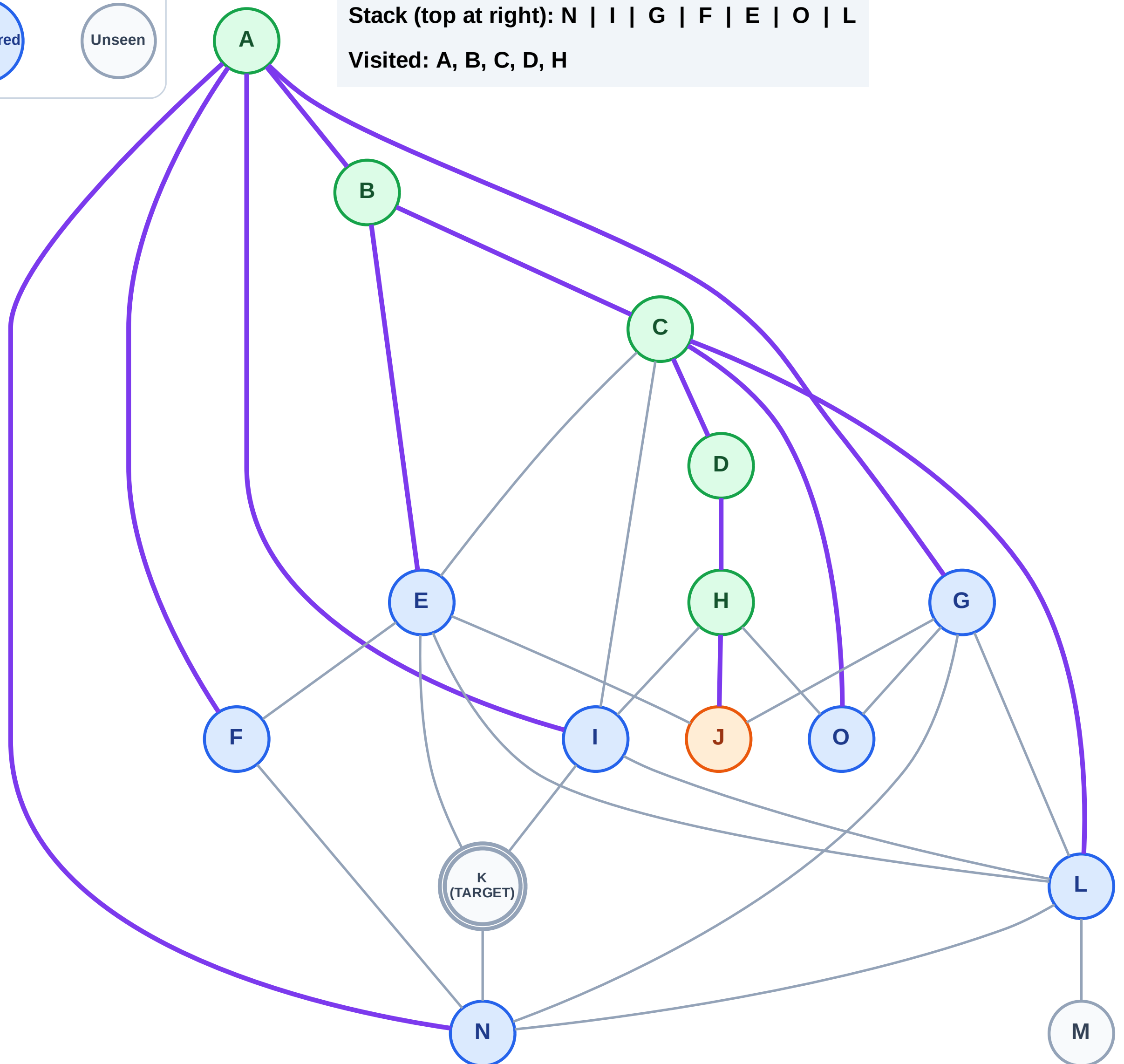
Stack (top at right): N | I | G | F | E | O | L | J

Visited: A, B, C, D, H



Step 12: Process node J

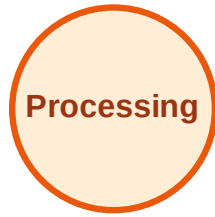
Step 12: Process node J



DFS Traversal

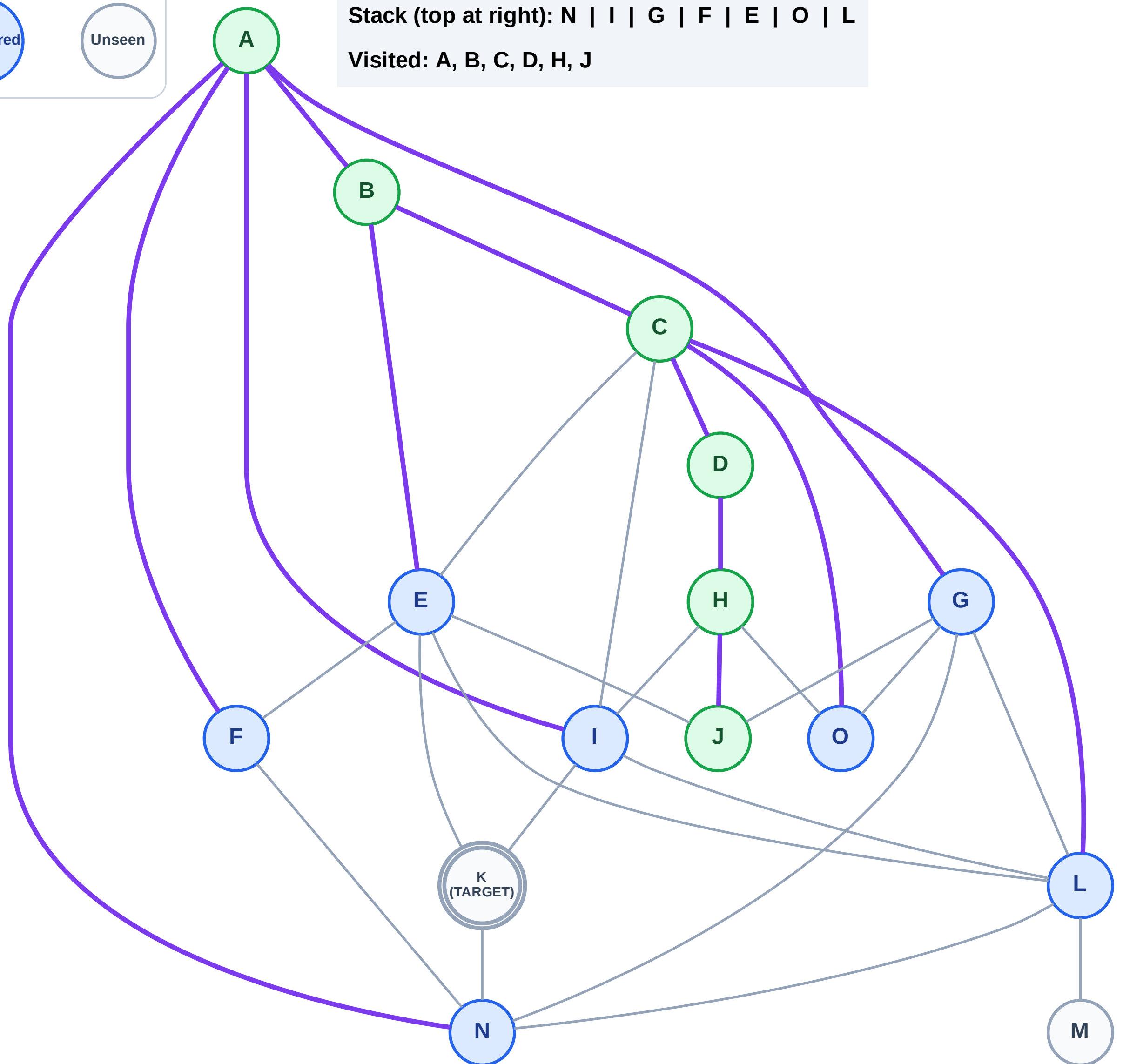
Step 13: No new neighbors from J

Legend



Stack (top at right): N | I | G | F | E | O | L

Visited: A, B, C, D, H, J



DFS Traversal

Step 14: Process node L

Legend

Complete

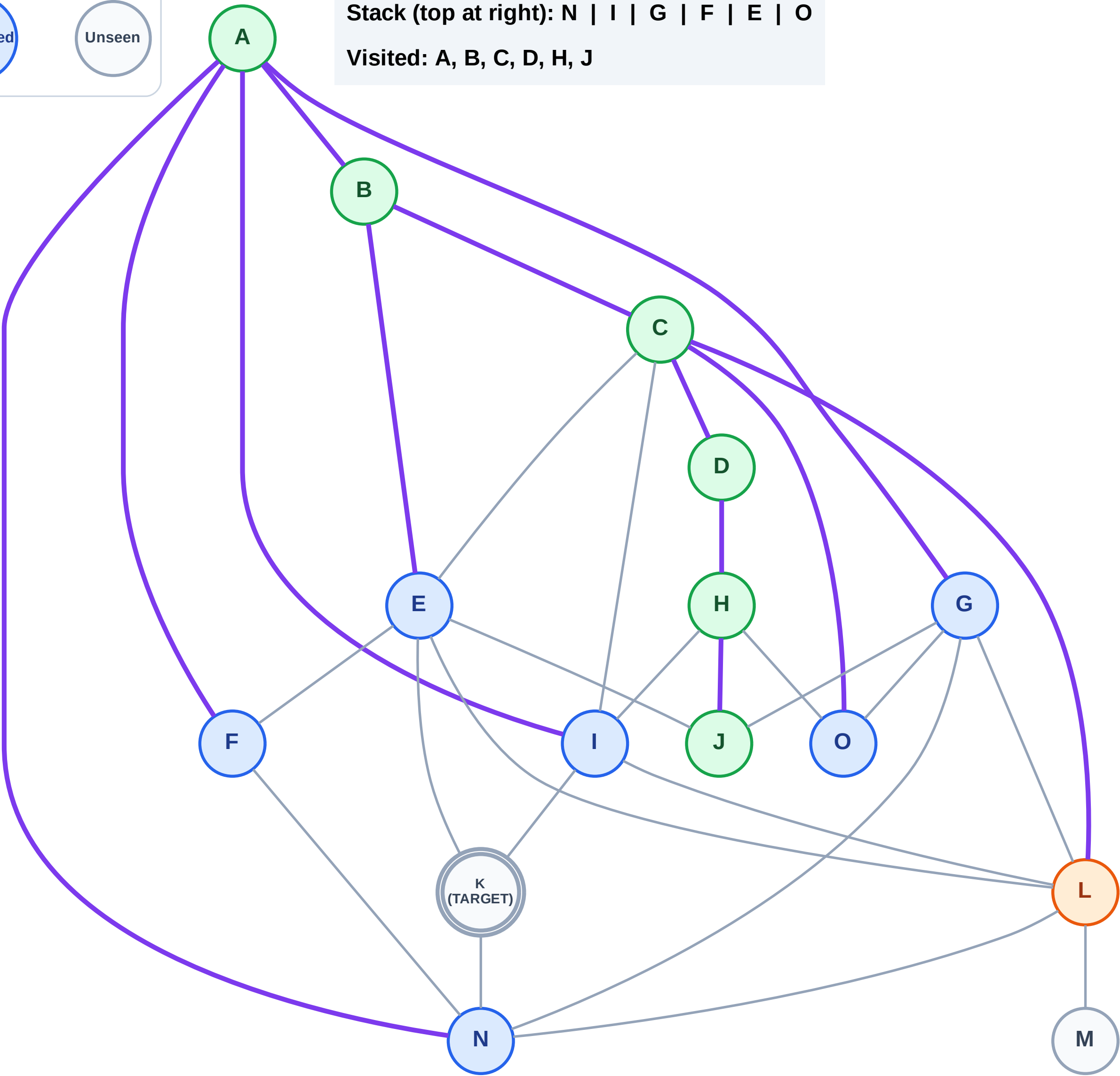
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F | E | O

Visited: A, B, C, D, H, J



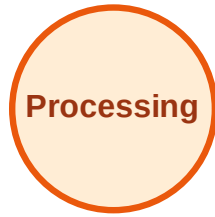
DFS Traversal

Step 15: Discover M from L

Legend



Complete



Processing



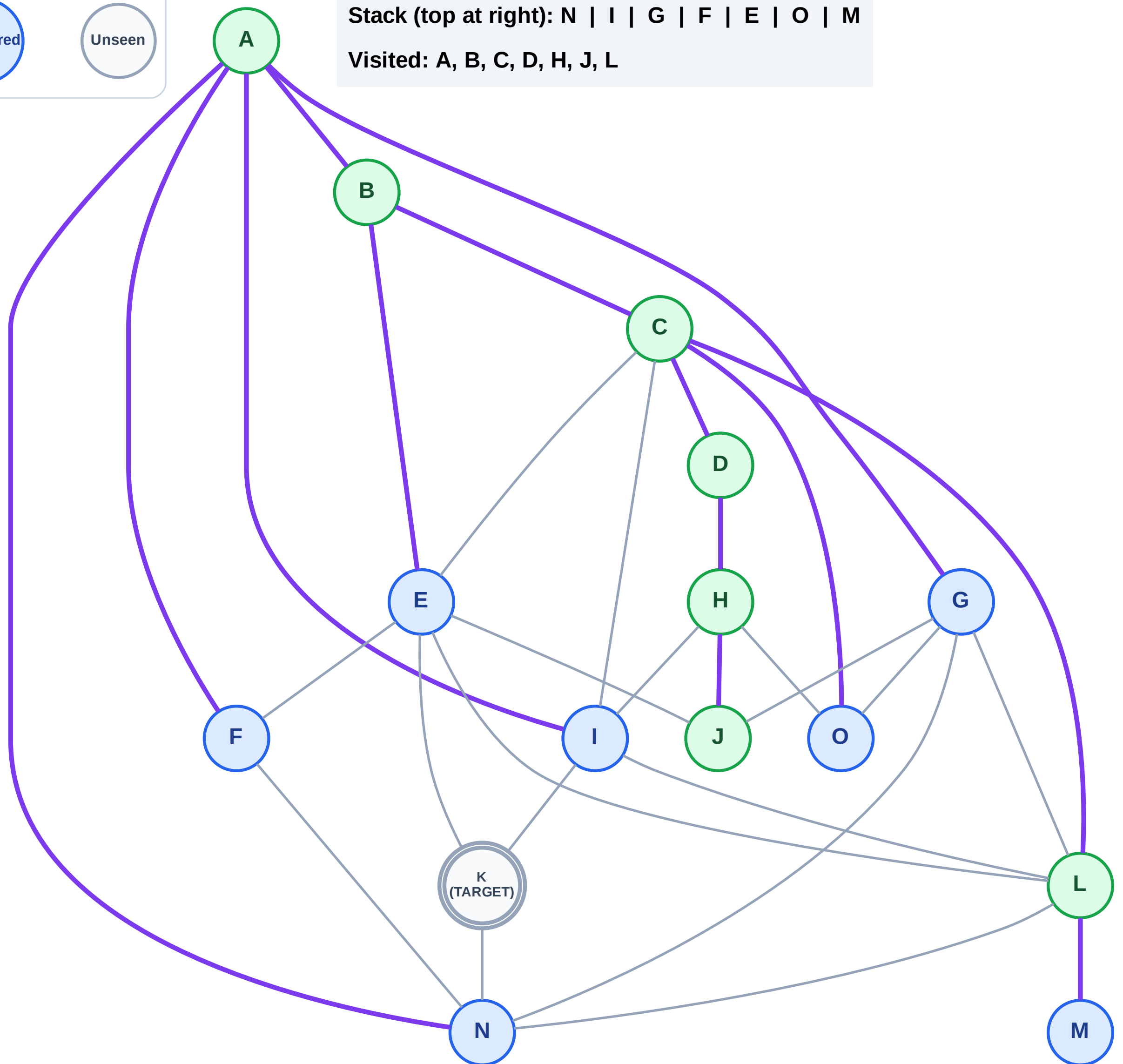
Discovered



Unseen

Stack (top at right): N | I | G | F | E | O | M

Visited: A, B, C, D, H, J, L



DFS Traversal

Step 16: Process node M

Legend

Complete

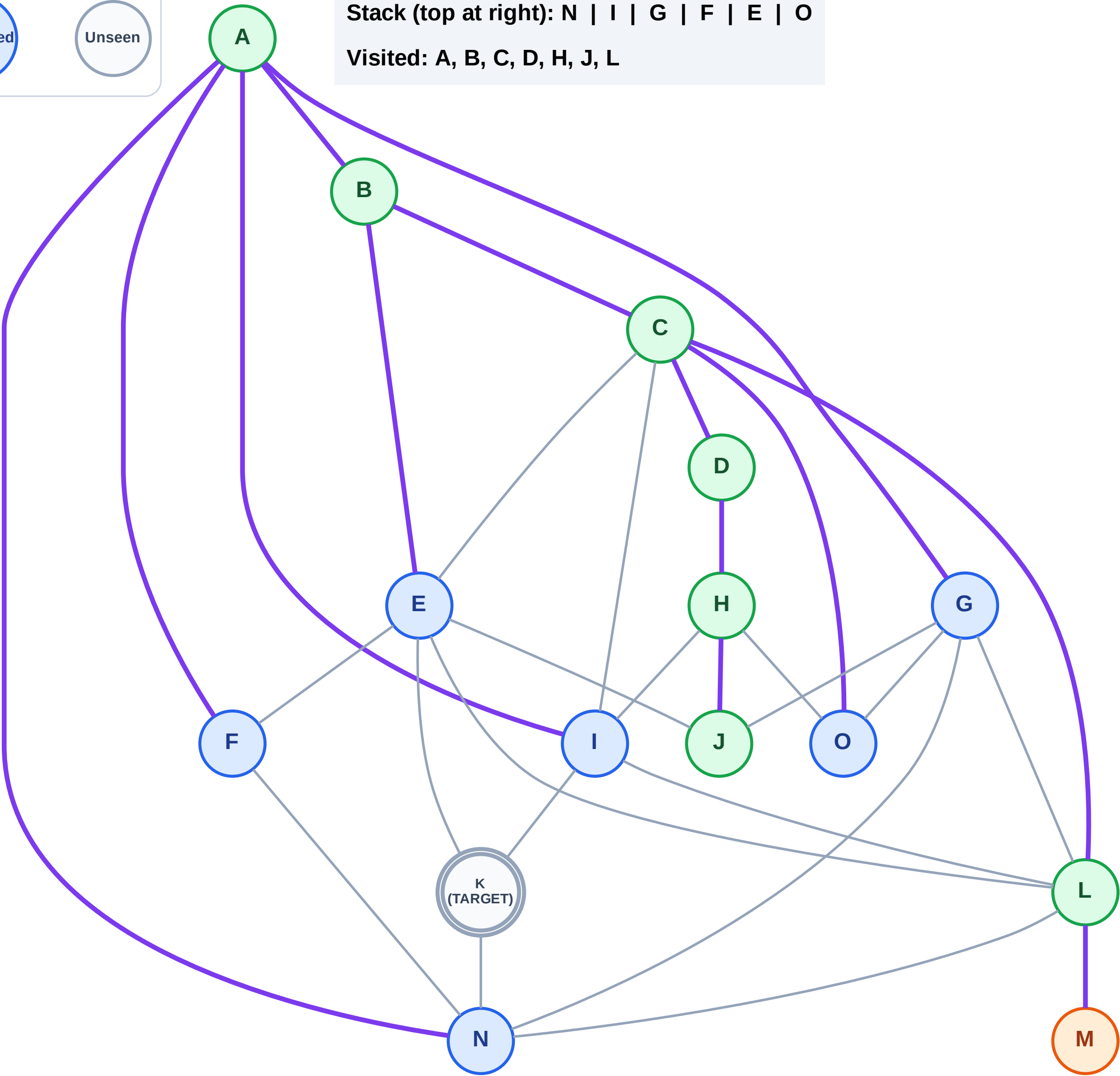
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F | E | O

Visited: A, B, C, D, H, J, L



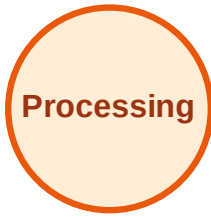
DFS Traversal

Step 17: No new neighbors from M

Legend



Complete



Processing



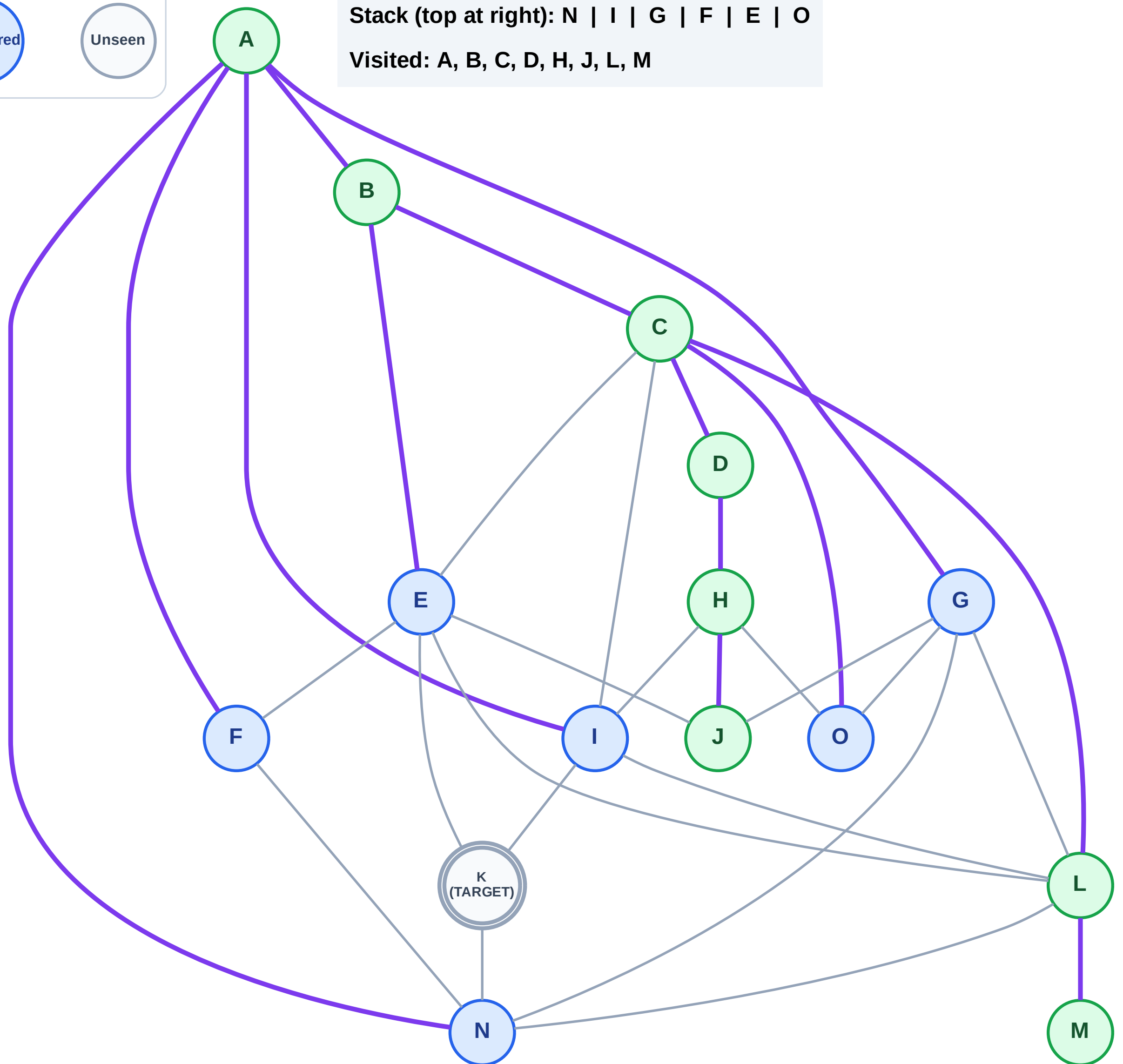
Discovered



Unseen

Stack (top at right): N | I | G | F | E | O

Visited: A, B, C, D, H, J, L, M



DFS Traversal

Step 18: Process node O

Legend

Complete

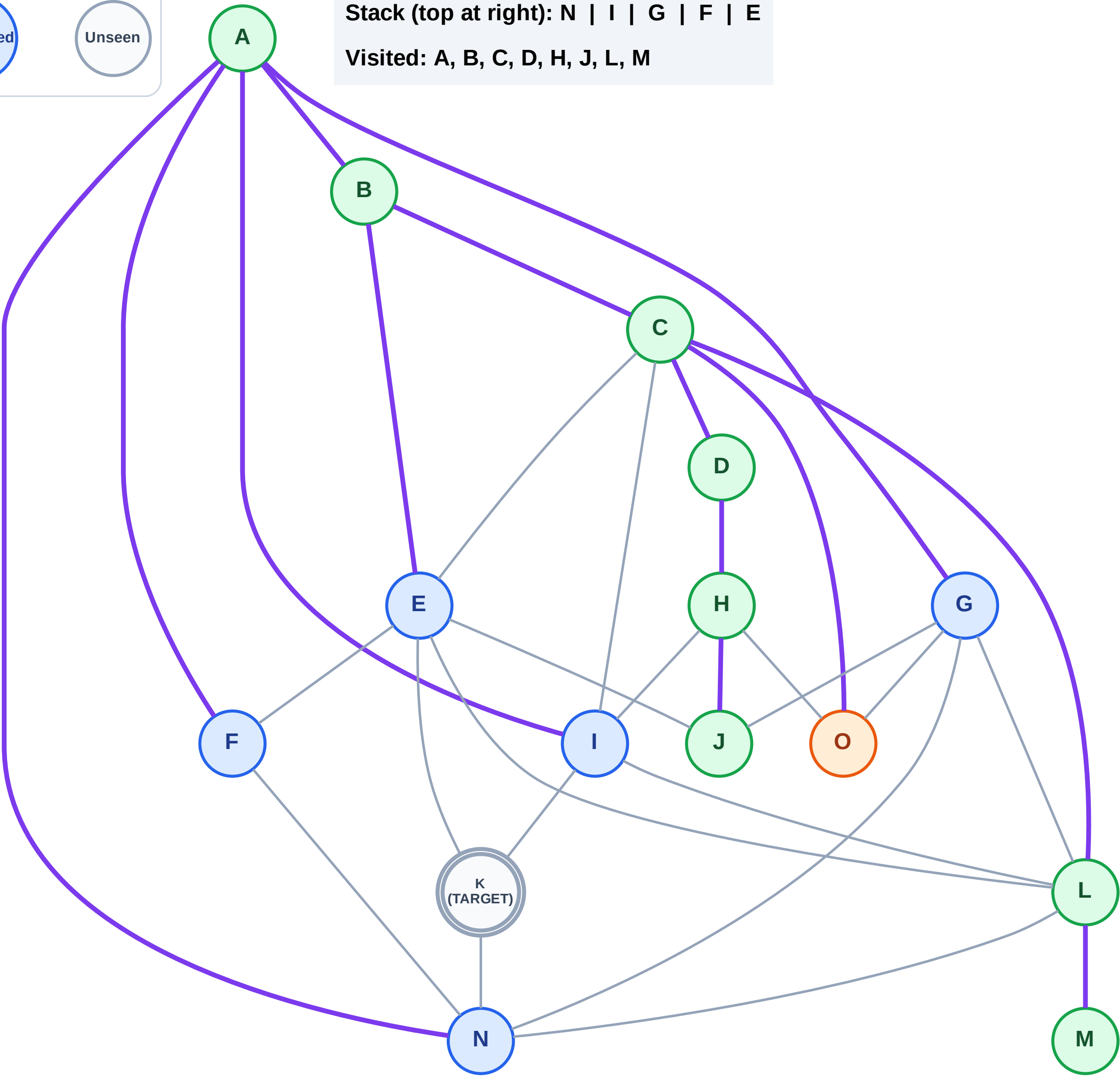
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F | E

Visited: A, B, C, D, H, J, L, M



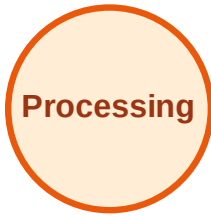
DFS Traversal

Step 19: No new neighbors from O

Legend



Complete



Processing



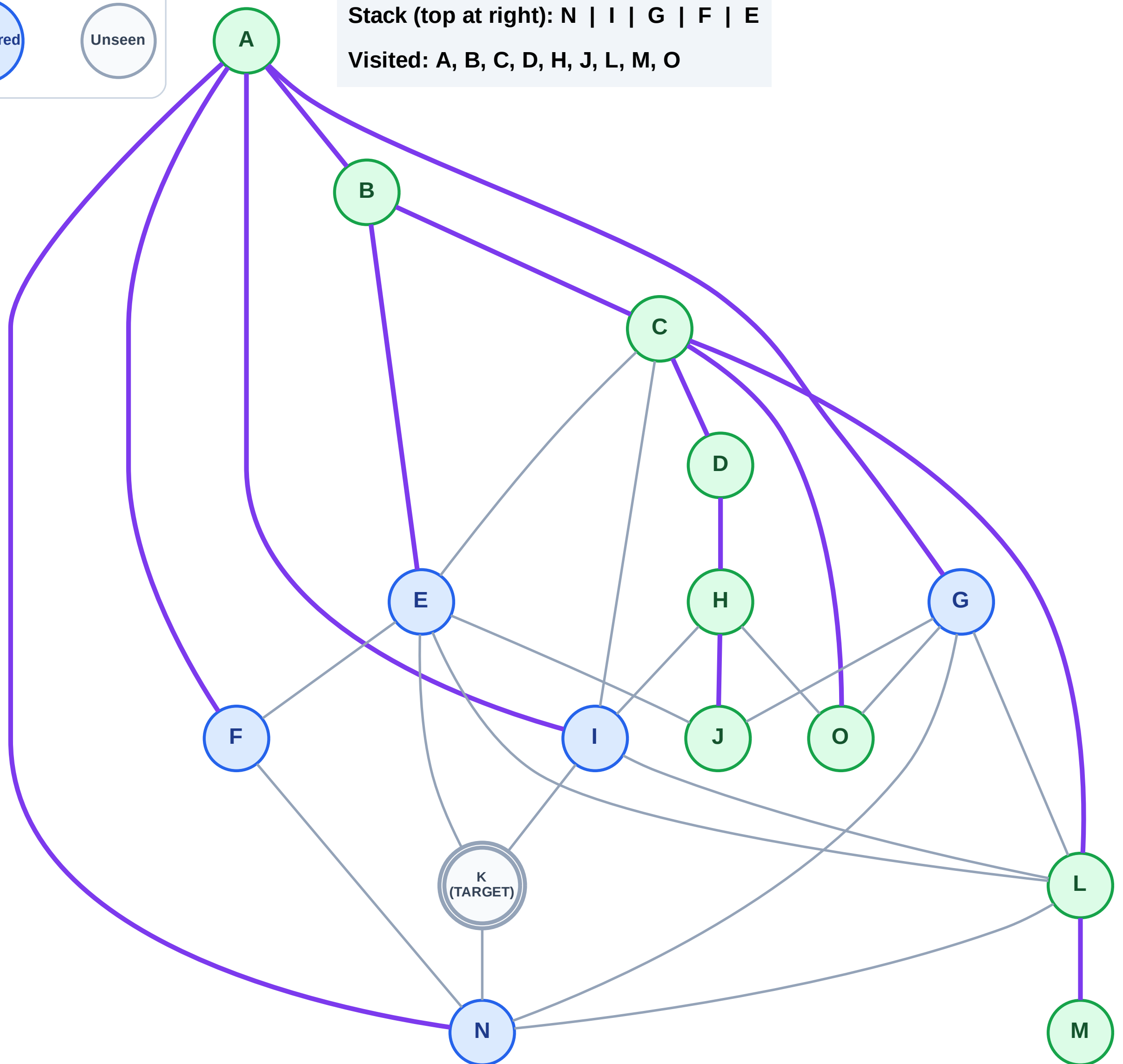
Discovered



Unseen

Stack (top at right): N | I | G | F | E

Visited: A, B, C, D, H, J, L, M, O



DFS Traversal

Step 20: Process node E

Legend

Complete

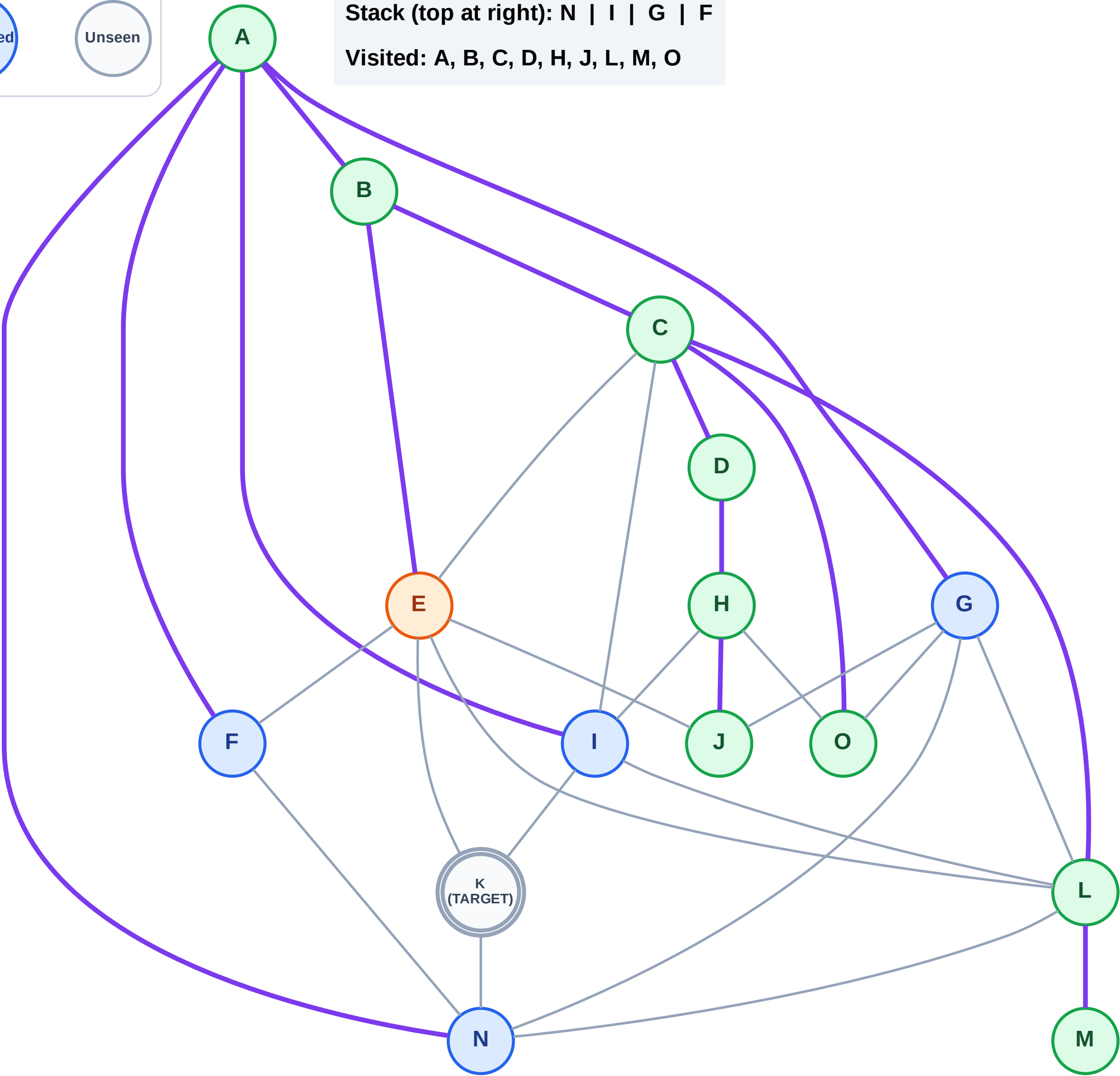
Processing

Discovered

Unseen

Stack (top at right): N | I | G | F

Visited: A, B, C, D, H, J, L, M, O



DFS Traversal

Step 21: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | G | F
Visited: A, B, C, D, E, H, J, K, L, M, O

